

An Introduction to Learning in Structured Domains & Learning on Graphs

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Work in Progress , please do not circulate!



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**Computational Intelligence
& Machine Learning Group**

ToC



- **Introduction**

- Motivations
- Historical notes



- **Main Course**

- Introduction to Deep Graph Networks
- Open Issues in Deep Graph Learning



- **Our Recent Proposals**

- Overview of some [advancement in my lab](#)
(to open new collaborations among us)
- Efficiency, Expressiveness, Graph XAI, Extensions

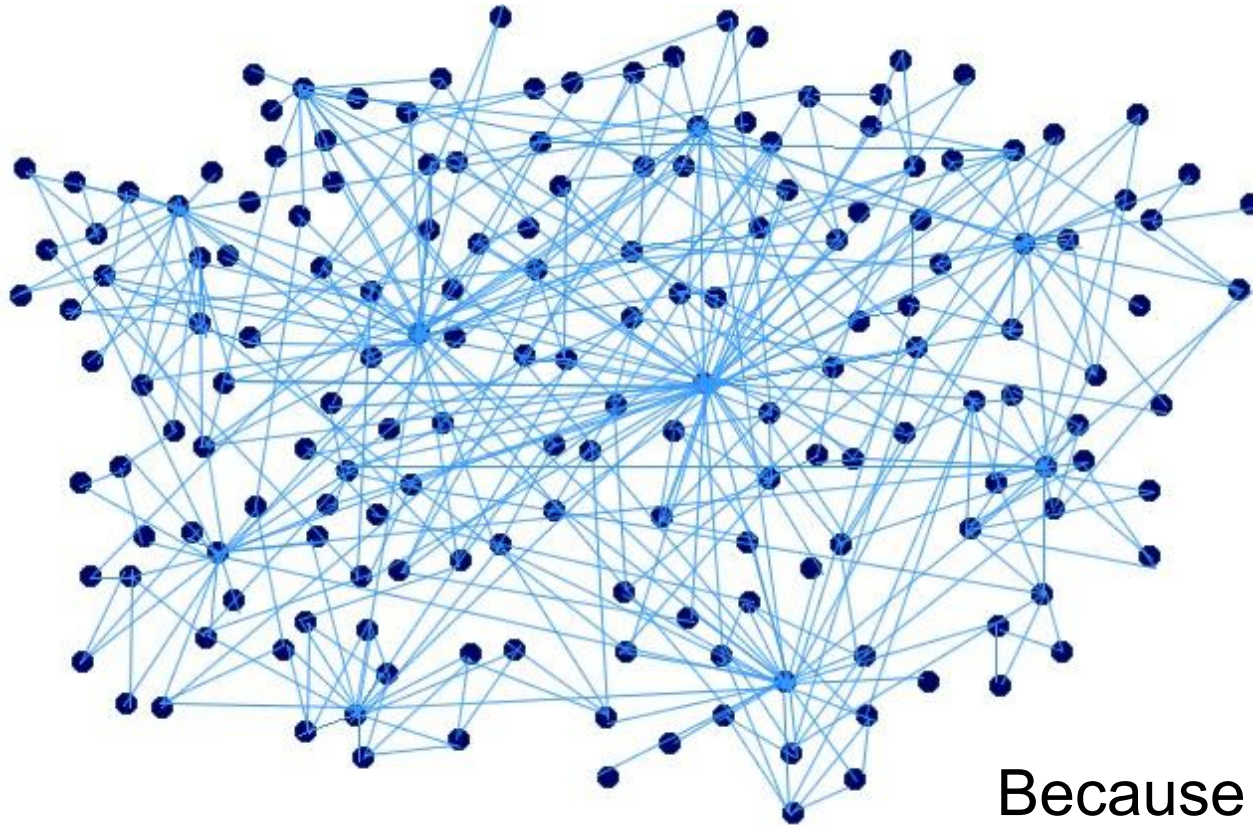


- **Other Approaches** (Kernels)

- **Perspectives & Conclusions**

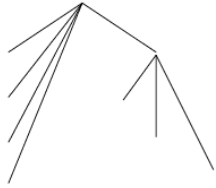
Overview & Motivations

Why Structured/Graph Data?

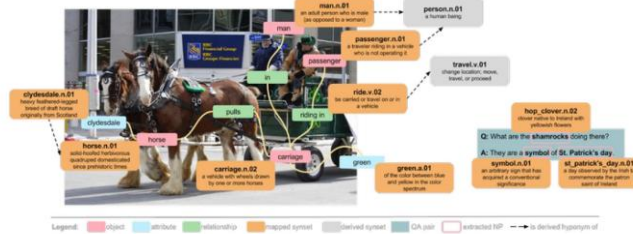


Because data have relationships

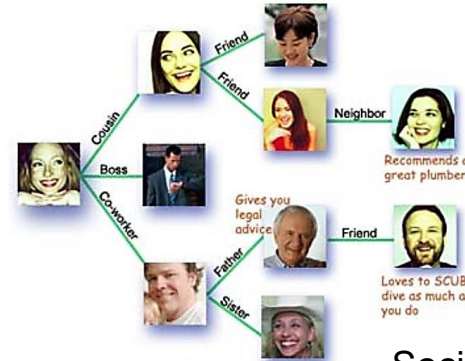
Examples: Graphs & Networks



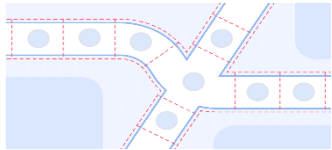
Abstraction of images



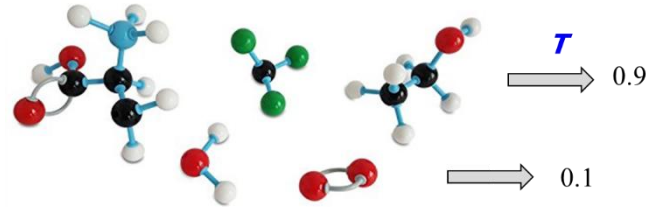
Scene understanding networks



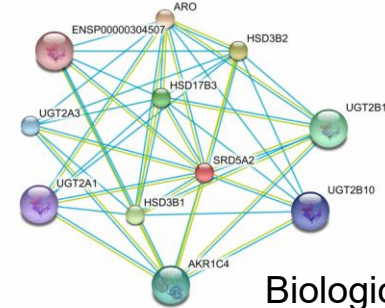
Social networks



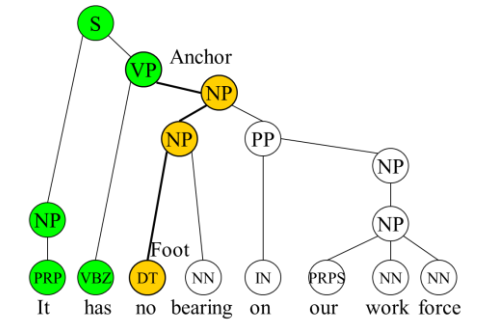
Transportation networks
(traffic predictions on G.maps@DeepMind)



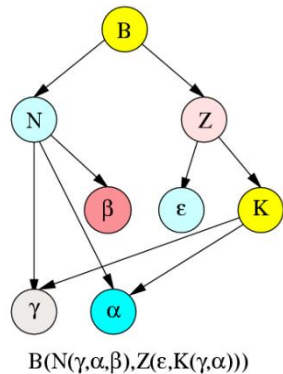
Small molecules (drug design)



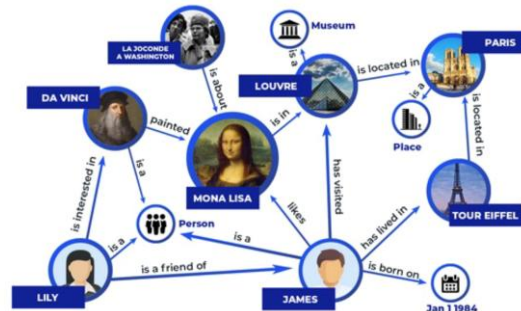
Biological networks
(proteins)



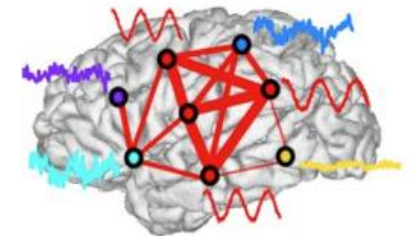
Language parsing



Logic terms

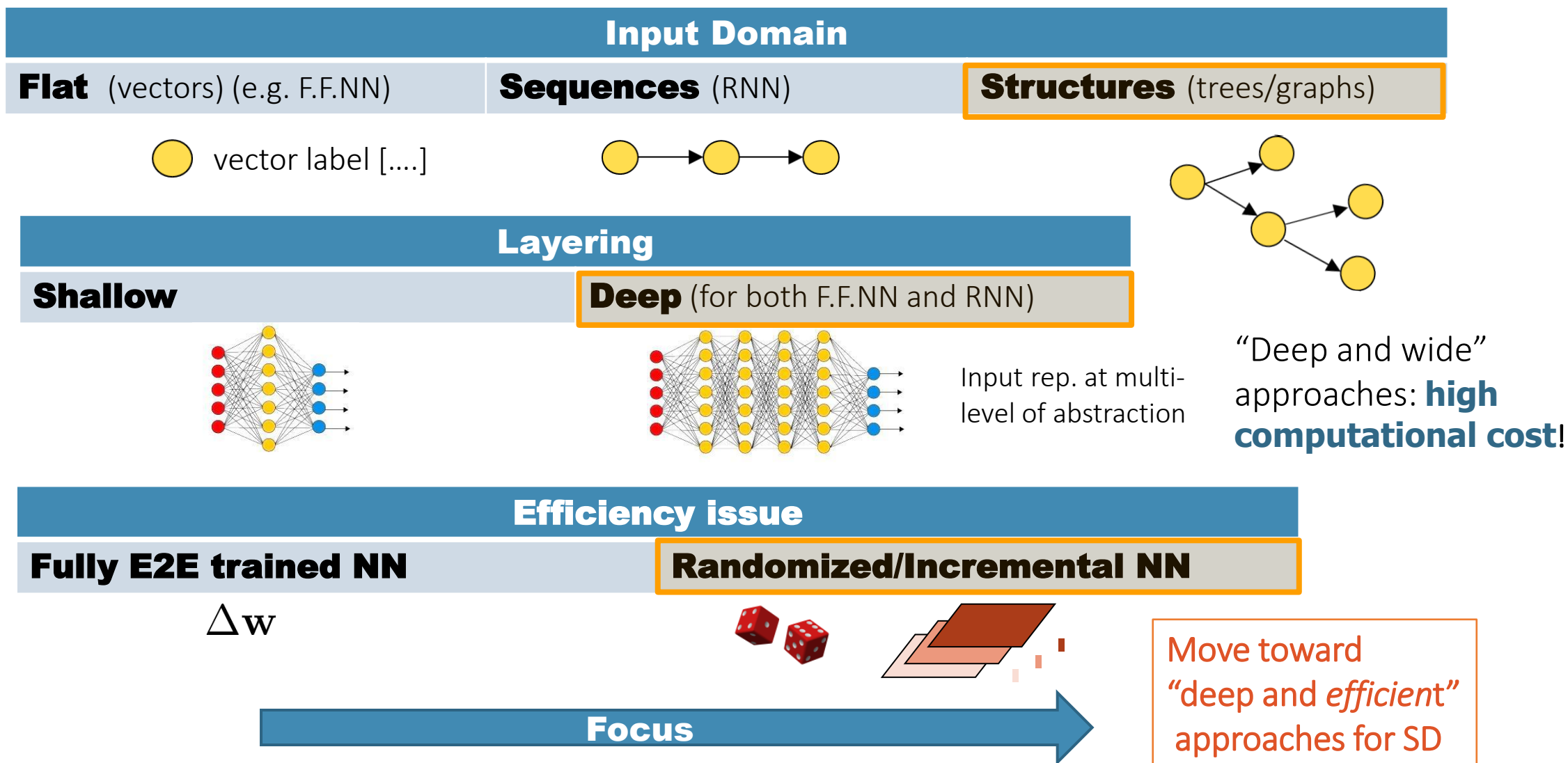


Knowledge graphs



Brain (networks) modeling

The scenario, terms (and trends)



The Question:



Why *learning on graphs* is attracting more and more research and applications?



- Many **SOLUTIONS** to previously difficult applicative problems, not respecting the *real nature* of the data (the relationships)
 - Instead of moving *data to models* we move *models to data*
- Many **NEW RESEARCH** problems arise when the domain is changed from flat to structured



Historical Notes

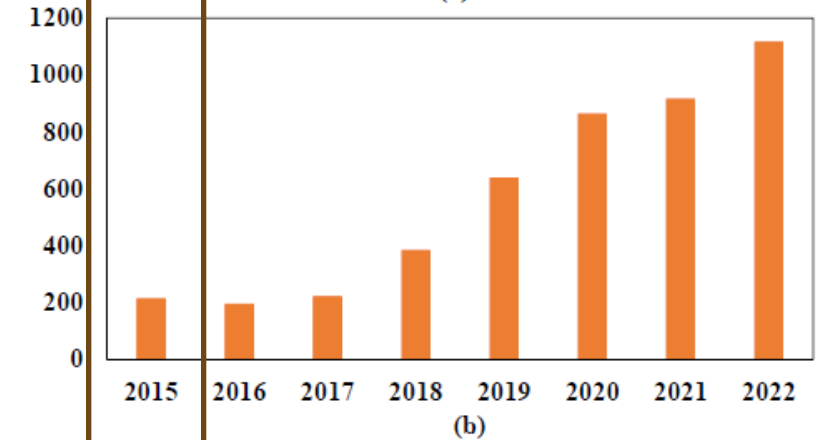
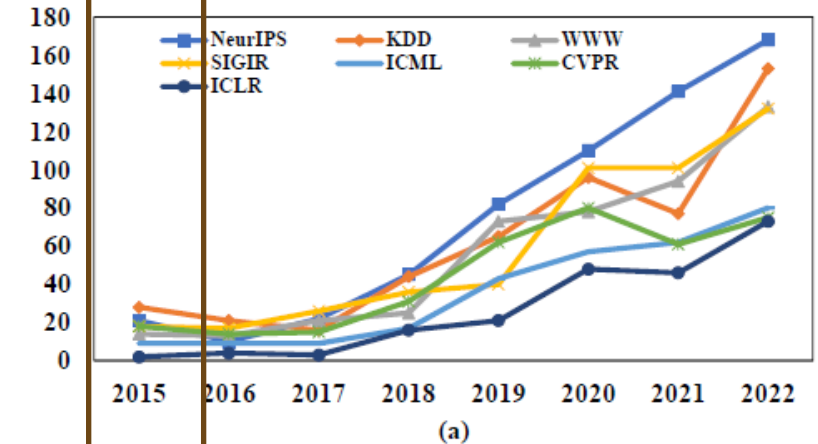
DGN: Evolutions

- Not surprisingly, nowadays, a world-wide “exploding” area:
 - Research literature is rapidly growing
 - And concrete real-world applications

Number of GL publications in recent years **on top venues**, including NeurIPS, KDD, WWW, SIGIR, ICML, CVPR and ICLR. Figures (a) and (b) depict the number of GL publications at top conferences and the **total number**, respectively

- Including different terminologies and perspectives, e.g.:
 - *Graph Representation Learning*,
 - *Learning on/for/with graphs*,
 - *Deep learning for graphs*
 - **Deep graph networks (DGN) framework** [Neural Netws. 2020]
 - ...

We will see neural models: Neural networks for graphs / Graph Neural Networks (GNN) / Graph Convolutional networks (GCN)/ ...



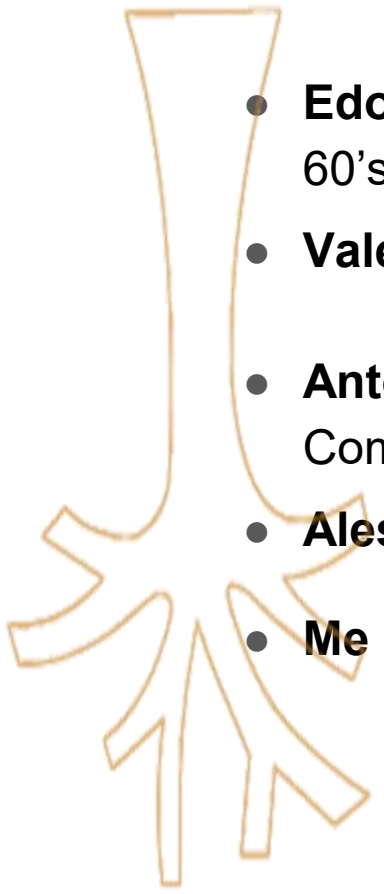
?

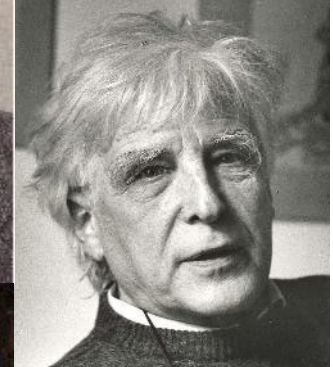
arxiv.org/abs/2212.08966

DGN: Origins, a Premise



The deep root and the path (timeline)

- 
- **Edoardo Renato Caianello** (Napoli, Italy), 60's: Neural networks, Cybernetics
 - **Valentino von Braitenberg**: Biocybernetics
 - **Antonina Starita**: Pisa "Neurolab", 80's: Computational Intelligence and Bioengineering
 - **Alessandro Sperduti** (Pisa & Padova), 90's
 - **Me ;-)**

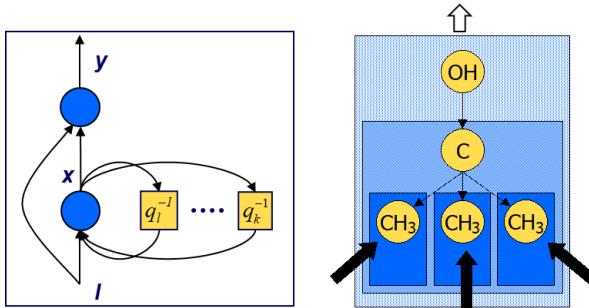


DGN: Origins (the roots, ... and the trees)

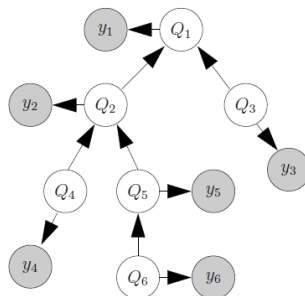


Origins of NNs for **Structured Data**, mainly an Italian history:

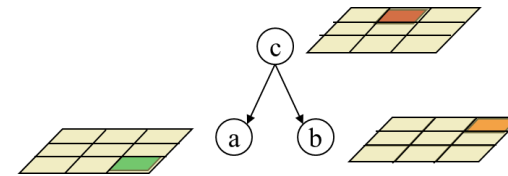
- Started with works on Trees, DPAGs etc. by *Recursive Neural Networks* (A. Sperduti, A. Starita,, P. Frasconi, M. Gori, B. Hammer, A. Micheli, M. Bianchini, F. Scarselli, M. Hagenbuchner, ...) since **end of 90's**, continuing for more than 25 years developing a large **family of models**, e.g.



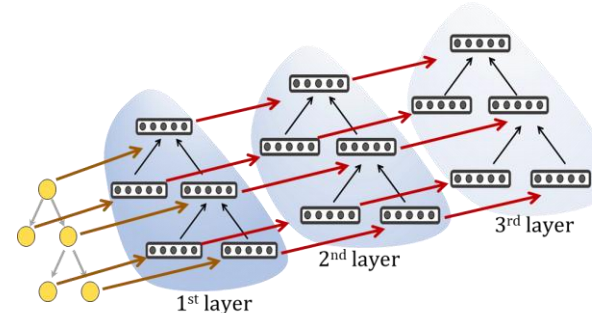
Recursive Neural Networks (1997-2000)



Bottom-up Hidden Tree Markov Models (2012)



Self-Organizing Map for Structures (2003-2005)



Tree ESN and Deep RC for Trees (2010-2018)

Origins of NNs for Structured Data (Trees/DPAGs/...): some references



- A. Sperduti, A. Starita. Supervised Neural Networks for the Classification of Structures, IEEE Transactions on Neural Networks. Vol. 8, n. 3, pp. 714-735, 1997.
- P. Frasconi, M. Gori, and A. Sperduti. **A General Framework for Adaptive Processing of Data Structures**, IEEE Transactions on Neural Networks. Vol. 9, No. 5, pp. 768-786, 1998.
- A.M. Bianucci, A. Micheli, A. Sperduti, A. Starita. **Application of Cascade Correlation Networks for Structures to Chemistry**, Applied Intelligence Journal (Kluwer Academic Publishers), Special Issue on "Neural Networks and Structured Knowledge" Vol. 12 (1/2): 117-146, 2000.
- A. Micheli, D. Sona, A. Sperduti. Contextual Processing of Structured Data by Recursive Cascade Correlation. IEEE Transactions on Neural Networks. Vol. 15, n. 6, Pages 1396- 1410, 2004.
- B. Hammer, A. Micheli, and A. Sperduti.
Universal Approximation Capability of Cascade Correlation for Structures.
Neural Computation. Vol. 17, No. 5, Pages 1109-1159, (C) 2005 MIT press
- B. Hammer, A. Micheli, A. Sperduti, M. Strickert.
Recursive Self-organizing Network Models. Neural Networks, Elsevier Science. Volume 17, Issues 8-9, Pages 1061-1085, 2004.
- C. Gallicchio, A. Micheli. **Tree Echo State Networks**, Neurocomputing, volume 101, pag. 319-337, 2013.
- C. Gallicchio, A. Micheli, **Deep** Reservoir Neural Networks for Trees. *Information Sciences* 480, 174-193, 2019

DGN: Origins



20th anniversary!

- Moved to **general graphs** (pioneered in [Pisa](#) and [Siena](#), Italy, **2005-2009**):
 1. **Recursive NN: GNN** [1], GraphESN [3] (see later)
 2. **Layering/Convolutional: NN4Gs** [2] in constructive version, and then continued as **Graph Convolutional NN (GCN)**
 - A line of research attracting more and more world-wide research since 2015...
- Deep Graph Networks (DGN) framework [4]

[1] F. Scarselli, M. Gori, A.C.Tsoi, M. Hagenbuchner, G. Monfardini. **The graph neural network model**, *IEEE Transactions on Neural Networks*, 20(1): 61–80, **2009**.

[2] A. Micheli. **Neural network for graphs: a contextual constructive approach**, *IEEE Transactions on Neural Networks*, 20 (3): 498-511, **2009**.

[3] C. Gallicchio, A. Micheli. **Graph Echo State Networks**, *IJCNN*, 2010.

[4] D.Bacciu, F. Errica, A. Micheli, M. Podda. A gentle introduction to deep learning for graphs, *Neural Netws.* 2020



IEEE TRANSACTIONS ON NEURAL NETWORKS

498

IEEE TRANSACTIONS ON NEURAL NETWORKS, VOL. 20, NO. 3, MARCH 2009

Neural Network for Graphs: A Contextual Constructive Approach

Alessio Micheli, Member, IEEE

Abstract—This paper presents a new approach for learning in structured domains (SDs) using a constructive neural network for graphs (NN4G). The new model allows the extension of the input domain for supervised neural networks to a general class of graphs including both acyclic/cyclic, directed/undirected labeled graphs. In particular, the model can realize adaptive contextual transductions, learning the mapping from graphs for both classification and regression tasks. In contrast to previous neural networks for structures that had a recursive dynamics, NN4G is based on a constructive feedforward architecture with state variables that uses neurons with no feedback connections. The neurons are applied to the input graphs by a general traversal process that relaxes the constraints of previous approaches derived by the causality assumption over hierarchical input data. Moreover, the incremental approach eliminates the need to introduce cyclic dependencies in the definition of the system state variables. In the traversal process, the NN4G units exploit (local) contextual information of the graphs vertices. In spite of the simplicity of the approach, we show that, through the compositionality of the contextual information developed by the learning, the model can deal with contextual information that is incrementally extended according to the graphs topology. The effectiveness and the generality of the new approach are investigated by analyzing its theoretical properties and providing experimental results.

Index Terms—Cascade correlation, contextual transductions, graph patterns, learning in structured domains (SDs), neural networks for structured data, recursive neural networks.

I. INTRODUCTION

THE study of learning in structured domains (SDs) allows for a generalization of machine learning (ML) approaches to the treatment of complex data, offering both new impulses for theory and applications. In particular, extending the methodologies in order to deal with classes of data structures can be recognized as one of the most promising ways of increasing the possibility of successful new applications. Indeed, varying-size structures, e.g., sequences, trees, and graphs occur in many fields such as natural language processing (NLP), document and structured text analysis, image processing, object recognition, cheminformatics, and bioinformatics. Learning in SD raises theoretical issues considering first the need of new models for non-vectorial data, which must be able to directly represent relational data of variable size in their hypothesis space, up to the generalization of the ML basis induced by the extension of the

data domains. Indeed, various methodologies approach the topic starting from different paradigms in the machine learning, statistical relational learning (see, e.g., [1] and [2]), and relational data mining area (see, e.g., [3]), with the aim of extending the class of input data that are traditionally restricted to a flat-vector form. Often the terms “structured domain learning” (SDL) or “relational learning” are used to refer the topic.

In this paper, we focus on subsymbolic supervised learning (classification/regression) for discrete data structures. We are recently witnessing an increasing attention in this area to the development of various kernel-based approaches for structured data (e.g., see [4]–[8] for overviews). Classical kernel approaches rely on the definition of a proper similarity measure on structures (expressed by the kernel function). An open issue is the problem of the design of a similarity measure on structures effective for specific tasks and/or of general validity, which poses practical and theoretical limitations [9], [10]. The point at issue is the concept of adaptivity, i.e., the capability that allows the model in principle to learn from the training data the similarity measure on structures for the task at hand.

In this respect, neural network approaches based on adaptive processing of the structured input data constitute a powerful methodology in the area of subsymbolic approaches to SDL. In particular, advanced models such as recurrent neural networks for sequence domains and recursive neural networks (RNNs) for SDs have been introduced [11]–[13]. In the family of RNNs, constructive approaches have been introduced for sequences and structures as well, such as recurrent cascade correlation [14] and recursive cascade correlation (RCC) [12], [15], [16]. RNN models realize an adaptive processing (encoding) of recursive (hierarchical) data structures. A recursive traversal algorithm is used to process (encode) all the graph vertices, producing state variable values for each visited vertex.

The RNN approach is computationally characterized by the causality of the transition system [11]. The causality assumption imposes on a directed acyclic graph that the computation for a vertex depends only on the current vertex and vertices descending from it (see Section II-B). Such assumption is used by RNN (and RCC) to realize an encoding that matches the input hierarchical structures according to their topological order. However, such encoding process poses constraints on the transduction realized by the RNN models (see [17]) and on the classes of data. Indeed, sequences, rooted trees, or directed (positional) acyclic graphs (DAG¹ and DPAG) are subclasses of graphs for which a topological sort exists.²

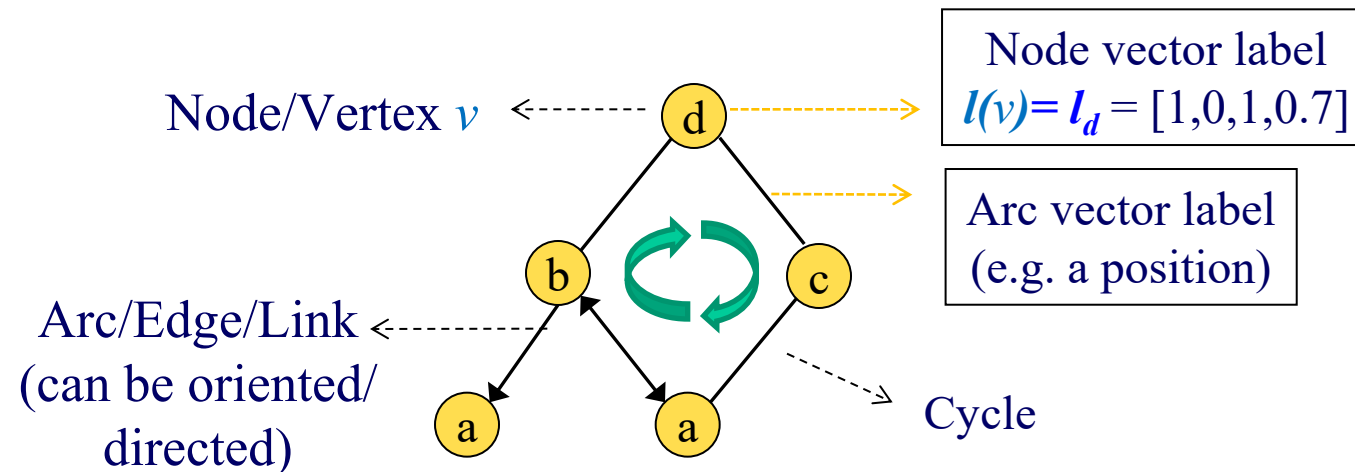
Manuscript received May 02, 2007; revised January 19, 2008 and July 29, 2008; accepted August 18, 2008. First published February 03, 2009; current version published February 27, 2009.
The author is with the Dipartimento di Informatica, Università di Pisa, 56127 Pisa, Italy (e-mail: alessio.micheli@unipi.it).

¹The treatment of DAG by RNN is introduced in [18].

Introduction to Deep Graph Networks

Our graphs (in the following)

Labeled graphs g



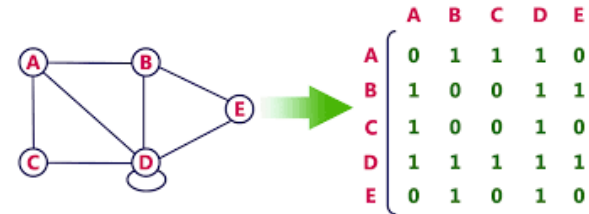
Call $\mathbf{A}$ the adjacency matrix of the graph

Graph Representations

The problem: there has been no systematic way (of general validity for any task) to extract features or metrics relations between examples for SD

- A representation learning instance (extended to SD)!
- **Features based** representations are incomplete (or strongly task-dependent, e.g. topological indexes)

- **Adjacent/incident matrix** representations (or other fixed-sizes representations). Issues:
 - Over-dim./incomplete (wasteful by padding/lose inf.)
 - Alignment among different graphs (!)
 - Topological order (it makes difficult the generalization)



- “*The ability to treat the proper **inherent nature of the input data** is the key feature for a successful application of the machine learning methodologies*”

Learning Models for SD

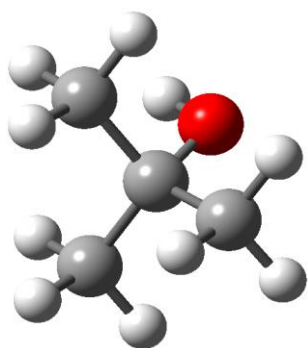
SD + ML = Learning in Structured Domains

Instead of moving *data to models*
(e.g. Graphs into vectors or trees into sequences,
with alignment problems, loose of information, etc.)
we move *models to data*

Example (graphs): Molecules and **Transductions**



- **The problem:** there has been no systematic way to extract features or metrics relations between examples for SD
- **Goal:** to learn a mapping between a structured information domain (SD) and a discrete or continuous space (*transduction*)



$$\text{Property/Activity} = \underset{\textcolor{blue}{T}}{\textcolor{blue}{T}}(\text{Structure})$$



Property Value (regression)
Toxic (yes/no) (classification)

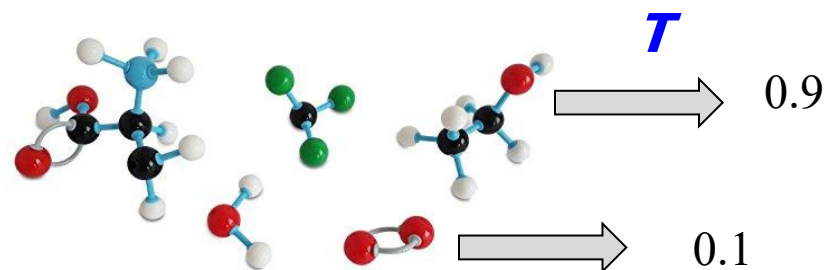
QSPR: Correlate chemical structure of molecules with their properties

Molecules are not vectors !
Molecules can be more naturally represented by varying size **structures**

Can we predict directly from structures ?

Learning a Transduction

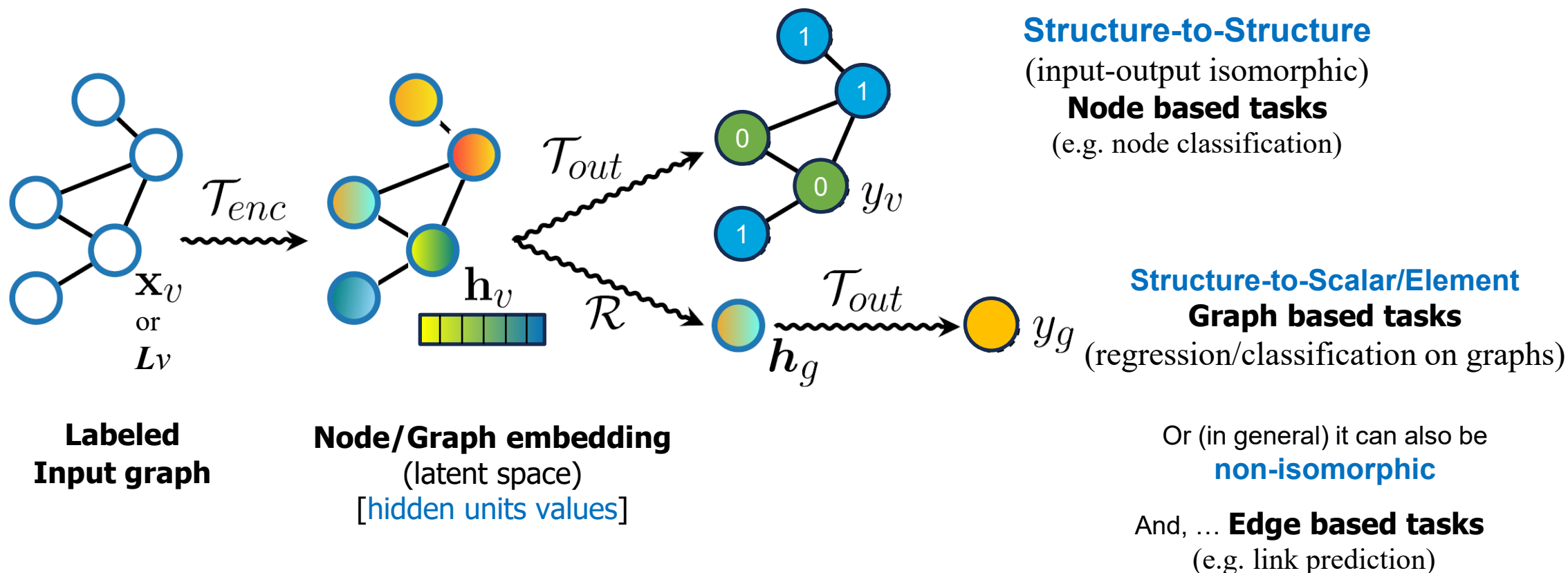
- **Goal:** to learn a mapping between a structured information domain (SD) and a discrete or continuous space (*transduction* T).
- Start with this problem: classify variable size graphs
 - For instance, classify different graphs (or regression task) starting from a training set of know couples as in the molecules example
- Given a set of examples $(graph_i, target_i)$
- Learn a hypothesis, i.e a mapping $T(graph)$



Tasks and Transductions on Graphs

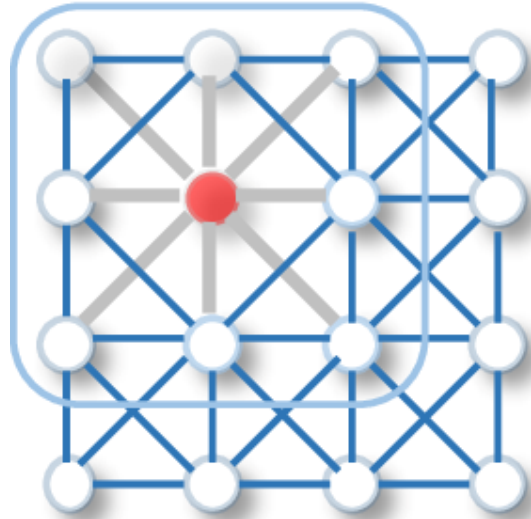


A transduction on graphs can be either



How DGNs work? Layering/ Convolutional approaches:

CNN and CGN both visit each node of the input data, but ...



CNN: Convolutional kernel can be applied on a 2D regular grid (image matrix)

- The CNN takes a weighted average of the pixel values in the neighbor window
- The **neighbor** of a vertex are **ordered** and have a fixed size



CNN cannot be immediately applied:

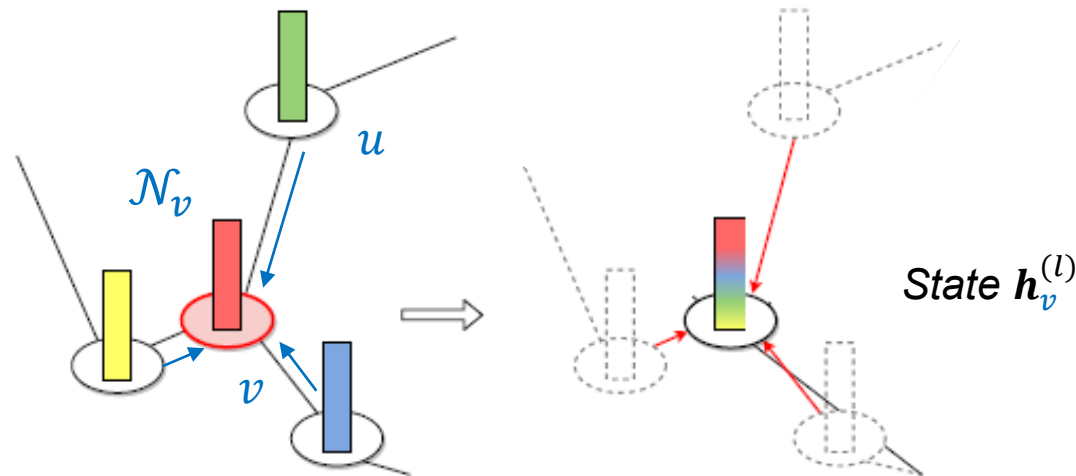
For **graphs** the **neighbors** of a vertex are **unordered and variable in size**

- We constrain the receptive field to **neighbors**, and we can exploit *weight sharing* to be permutation invariant so to *abstract from any ordering of the nodes*
- And **we extend from local neighbor through layering**: see next slides

How DGNs work? «Message Passing» Concept



$$\mathbf{h}_v^{(l)} = \text{AGG}_{\mathbf{W}^{(l)}}(\mathbf{L}_v, \mathbf{h}_v^{(l-1)}, \{\mathbf{h}_u^{(l-1)} : u \in \mathcal{N}(v)\}) \quad l = 1, \dots, L$$

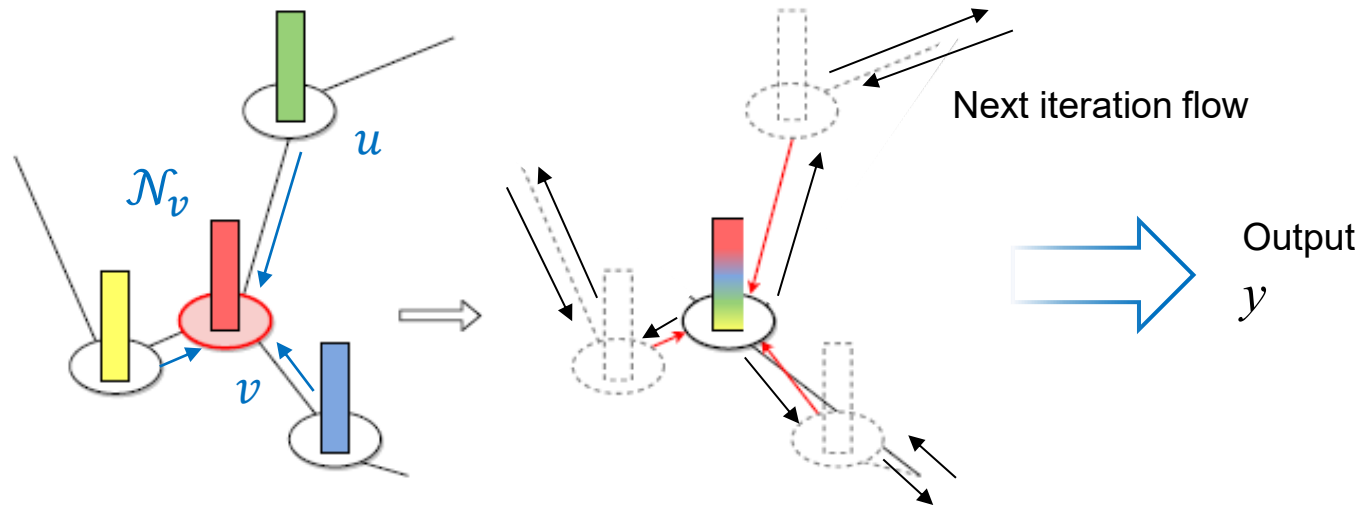


- Nodes “talk” *each other* to inform about themselves and to be aware of the surrounding structure, by *messages*
- We compute for each node a *State* $\mathbf{h}_v^{(l)}$ combining the *messages* from $\mathcal{N}_v$ by *neural networks* with free parameters $\mathbf{W}$, allowing to learn the node embedding

How DGNs work? «Message passing» Concept (II)

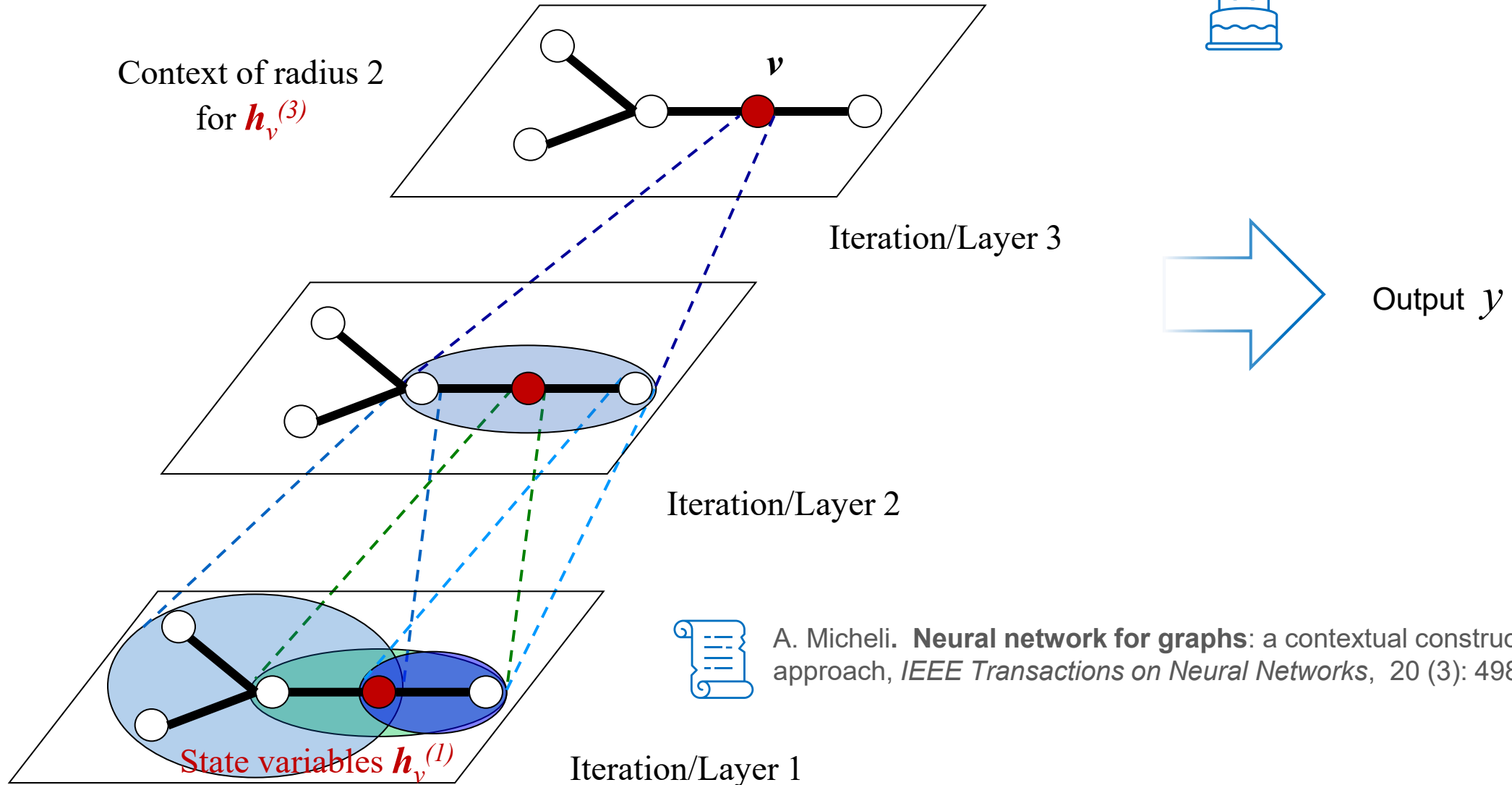


$$\mathbf{h}_v^{(l)} = \text{AGG}_{\mathbf{W}^{(l)}}(\mathbf{L}_v, \mathbf{h}_v^{(l-1)}, \{\mathbf{h}_u^{(l-1)} : u \in \mathcal{N}(v)\}) \quad l = 1, \dots, L$$



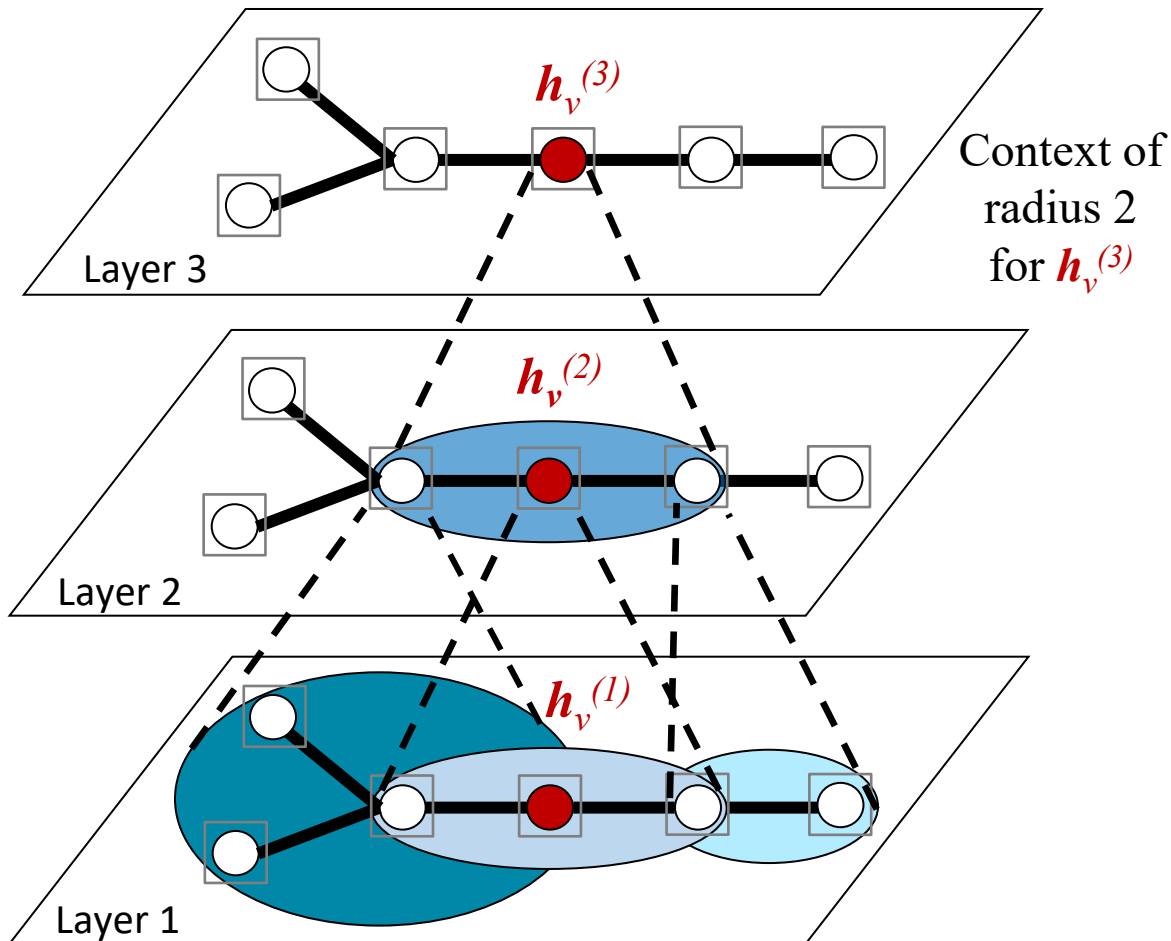
- The state does *not* depend only on local *neighborhood* $\mathcal{N}_v$: **Iterations** of this MP mechanism allows the **spread/diffusion of information** for all the nodes to all the others
- Finally, an **output** can be emitted **for each node** or **for the whole graph** through a permutation-invariant aggregation of node representations, also called *global pooling* (e.g. a sum or a mean operator)

«Message Passing» Concept: Iterations in GCN (the movie)



A. Micheli. **Neural network for graphs**: a contextual constructive approach, *IEEE Transactions on Neural Networks*, 20 (3): 498-511, **2009**

GCN & NN4G (2005-2009) Summary



Iterations l of MP are new layers

- For each node we compose the context/
receptive fields through layers (compositionality)
- Depth (DL) is functional to the context growth
→ DGNs are *inherently deep*
- Training the model we learn how to represent the graph:
 - E2E training for general GCNs
 - Layer-wise (incremental) training for NN4G
 - (automatic # of layers, no gradient vanish issues, etc)



A. Micheli. **Neural network for graphs: a contextual constructive approach**, *IEEE Transactions on Neural Networks*, 20 (3): 498-511, 2009

Message Passing: Details

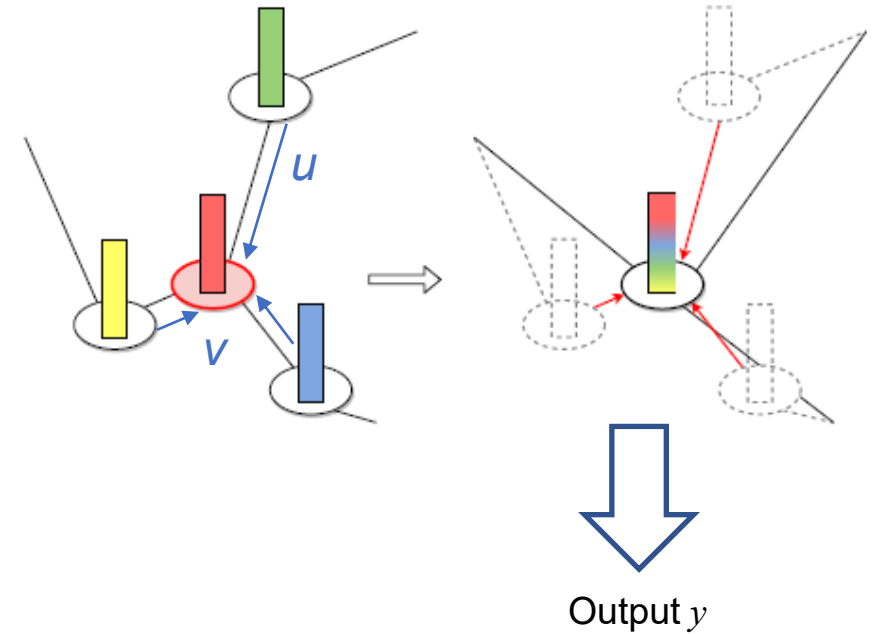


- Iteration (over l) of

Message

$$\mathbf{h}_v^{(l)} = \text{AGG}_{\mathbf{W}^{(l)}}(\mathbf{L}_v, \mathbf{h}_v^{(l-1)}, \{\mathbf{h}_u^{(l-1)} : u \in \mathcal{N}(v)\}) \quad l = 1, \dots, L$$

- $\text{AGG}_{\mathbf{W}^{(l)}}$ aggregates (propagates) **messages** from the neighbors nodes,
 - it includes a permutation invariant operator over the set of neighbors $\mathcal{N}$ (e.g. a sum),
 - and combinations of the NN units' activations in $\mathbf{h}$, where $\mathbf{W}$ are the free model parameters.
 $\mathbf{L}_v$ (or $\mathbf{x}_v$) are the node's labels.
 - $\mathcal{N}$ is according to the adjacency matrix $\mathbf{A}$ of the graph.
- Applying this computation to each node of the graph corresponds to a parallel, unordered **visit** of the graph at each iteration l



Message Passing: Implementations



$$\mathbf{h}_v^{(l)} = \text{AGG}_{\mathbf{W}^{(l)}}(\mathbf{L}_v, \mathbf{h}_v^{(l-1)}, \{\mathbf{h}_u^{(l-1)} : u \in \mathcal{N}(v)\}) \quad l = 1, \dots, L$$

- We have a large **flexibility** for implementing these operators, with many model variations ($\mathbf{A}$ is the adjacency matrix of the graph):
 - **GCN** [Kipf, Welling, ICLR 2017] $\mathbf{h}^{(l)} = \text{relu}(\hat{\mathbf{A}} \mathbf{h}^{(l-1)} \mathbf{W}^{(l)})$, $\hat{\mathbf{A}}$ degree normalized
 - **GraphConv** [Morris et al., AAAI 2019] $\mathbf{h}^{(l)} = f(\mathbf{W}_1^{(l)} \mathbf{h}^{(l-1)} + \mathbf{W}_2^{(l)} \mathbf{A} \mathbf{h}^{(l-1)})$
 - **GAT** [Veličković et al., ICLR 2018]: Attention-based aggregation (by $\alpha_{v,u}$)



pytorch-geometric

We named as **DGN** framework (including NN, Bayesian, etc) [Neural Netws. 2020]

NN4G (GCN): The Equations



Message

$$\mathbf{h}_v^{(l)} = \text{AGG}_{\mathbf{W}^{(l)}}(\mathbf{L}_v, \mathbf{h}_v^{(l-1)}, \{\mathbf{h}_u^{(l-1)} : u \in \mathcal{N}(v)\}) \quad l = 1, \dots, L$$

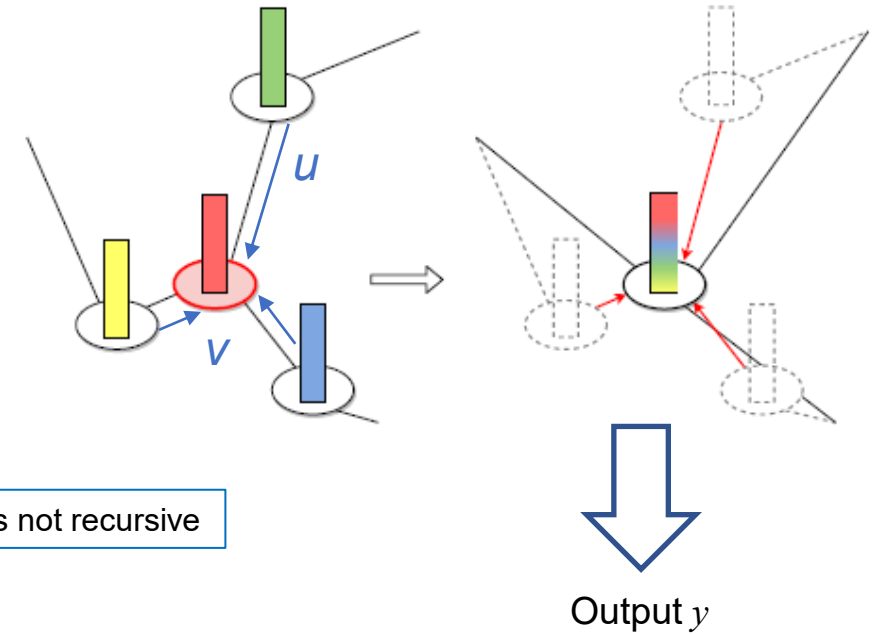
- NN4G computes a (one) state $\mathbf{h}_v^{(l)}$ for each layer and for each vertex v in this way:

$$\mathbf{h}_v^{(l)} = \begin{cases} h_v^{(1)} = f\left(\sum_{j=0}^{L^v} \bar{w}_{1j} L_j(v)\right) & \text{First layer } (l=1) \\ h_v^{(l)} = f\left(\sum_{j=0}^{L^v} \bar{w}_{lj} L_j(v) + \sum_{j=1}^{l-1} \hat{w}_{lj} \sum_{u \in \mathcal{N}(v)} h_u^{(j)}\right) & l = 2, \dots, N \end{cases}$$

Previous frozen layers: It is not recursive

Current Label of v

Context from the Neigh.s
from (all) previous layers



A. Micheli. **Neural network for graphs**: a contextual constructive approach, *IEEE Transactions on Neural Networks*, 20 (3): 498-511, **2009**

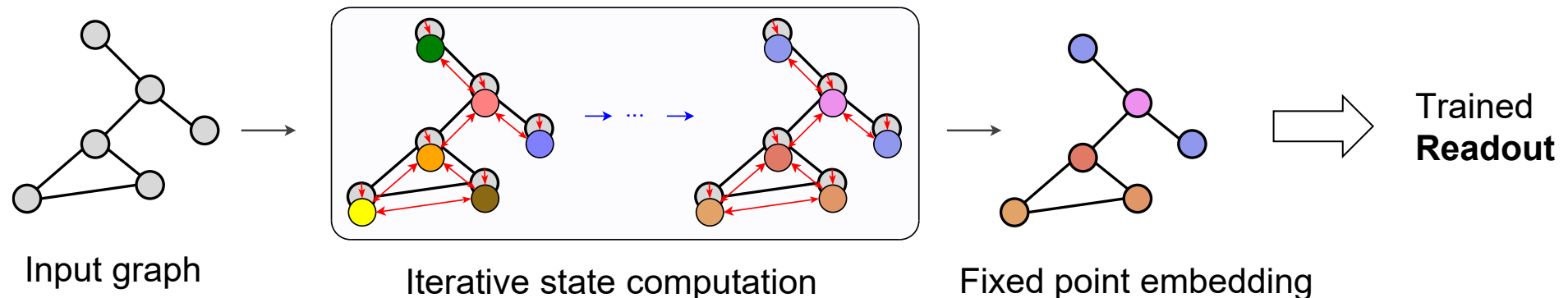
GNN-GESN (2005-2010)



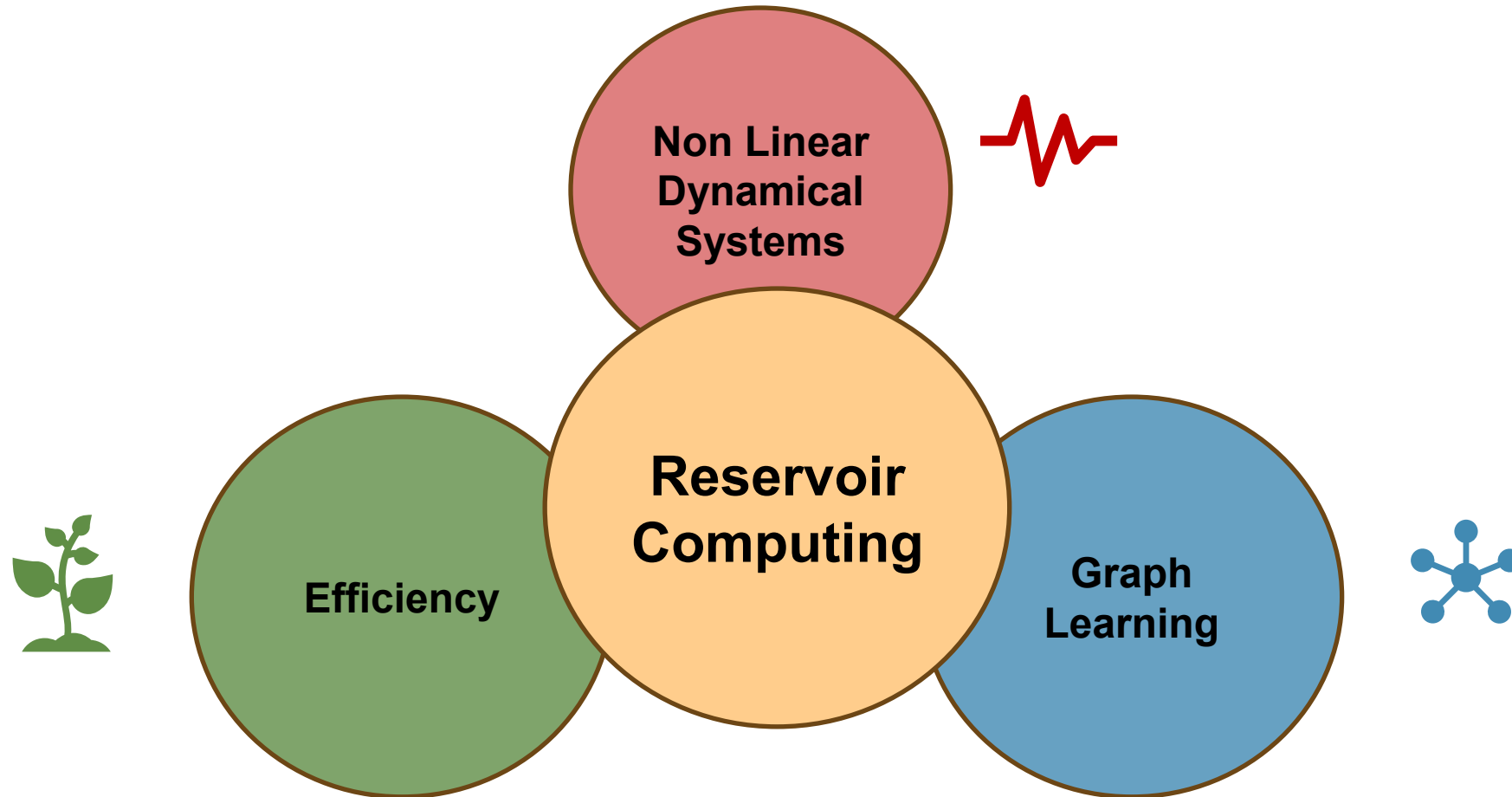
- For the *recursive approaches* (**GNN**, **GESN**) the diffusion process is similar, but l is the number of MP iterations within the same layer (or different layers with shared weights among layers)

$$h_v^{(l)} = \text{AGG}_{W^{(l)}}(L_v, h_v^{(l-1)}, \{h_u^{(l-1)} : u \in \mathcal{N}(v)\})$$

- Dynamical system**: typically requires convergence to a fixed point as graph embedding, constraining (MP) state dynamics to be contractive
 - GNN** with training for the embedding [Scarselli, et al., TNNLS 2009]
 - GESN** (extending **randomized** Echo State Networks for sequences): **untrained** message-passing function (but direct control of the dynamical system parameters) [Gallicchio, Micheli, IJCNN, 2010; AAAI 2020]



GraphESN: Reservoir Computing for Graphs



GraphESN: Reservoir Computing for Graphs



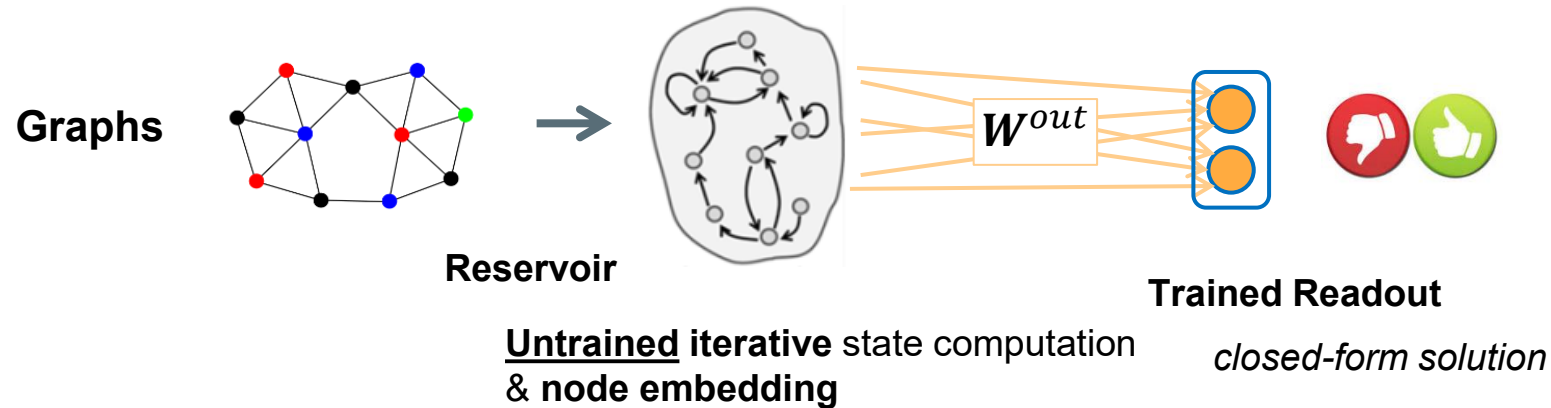
We can extend RC to **efficient** graph learning: GraphESN (GESN)

- Origins: Gallicchio, Micheli. *Graph echo state networks*, IJCNN 2010
- Deep: Gallicchio, Micheli. *Fast and Deep Graph Neural Networks*. AAAI 2020



Reservoir: Abstract randomized
encoder machine for graphs

- **No training** (No backprop)



GESN: The Equations



Message

$$\mathbf{h}_v^{(l)} = \text{AGG}_{\mathbf{W}^{(l)}}(\mathbf{L}_v, \mathbf{h}_v^{(l-1)}, \{\mathbf{h}_u^{(l-1)} : u \in \mathcal{N}(v)\}) \quad l = 1, \dots, L$$

GraphESN compute state variables (here a vector $\mathbf{h}$ of reservoir units) for each vertex v in this way:

GraphESN state transitions (recursive) for l **iterations** until convergence

$$\mathbf{h}_v^{(l)} = \tanh \left(\sum_{j=0}^{L^v} \bar{w}_{lj} L_j(v) + \sum_{u \in \mathcal{N}(v)} \hat{\mathbf{W}} \mathbf{h}_u^{(l-1)} \right)$$

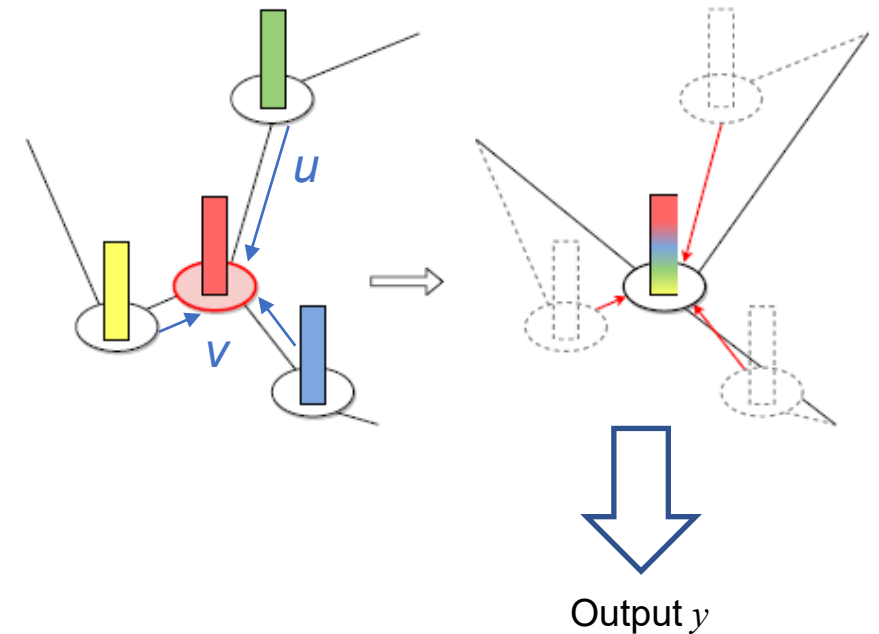
Think to a RNN, with initial null states for the Neighbors

Connectivity of the Reservoir

Context from the Neigh.s recursively from previous iteration

$$\mathbf{y}(\mathbf{g}) = \mathbf{W}_{out} \mathbf{X}(\mathbf{h}(\mathbf{g}))$$

Output for a graph $\mathbf{g}$ by *global pooling* $\mathbf{X}$ and then by a linear **Readout** trained by pseudo-inversion/Ridge Regression



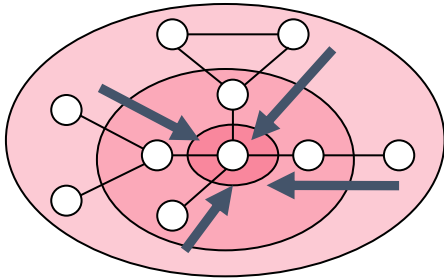
[Gallicchio, Micheli, IJCNN, 2010; AAAI 2020]

GESN: Iterations



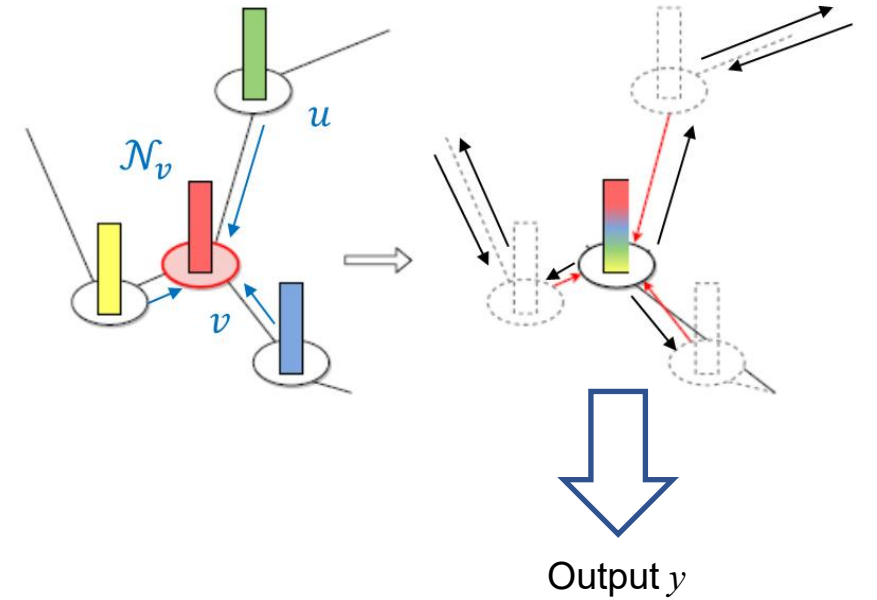
Message

$$\mathbf{h}_v^{(l)} = \text{AGG}_{\mathbf{W}^{(l)}}(\mathbf{L}_v, \mathbf{h}_v^{(l-1)}, \{\mathbf{h}_u^{(l-1)} : u \in \mathcal{N}(v)\}) \quad l = 1, \dots, L$$



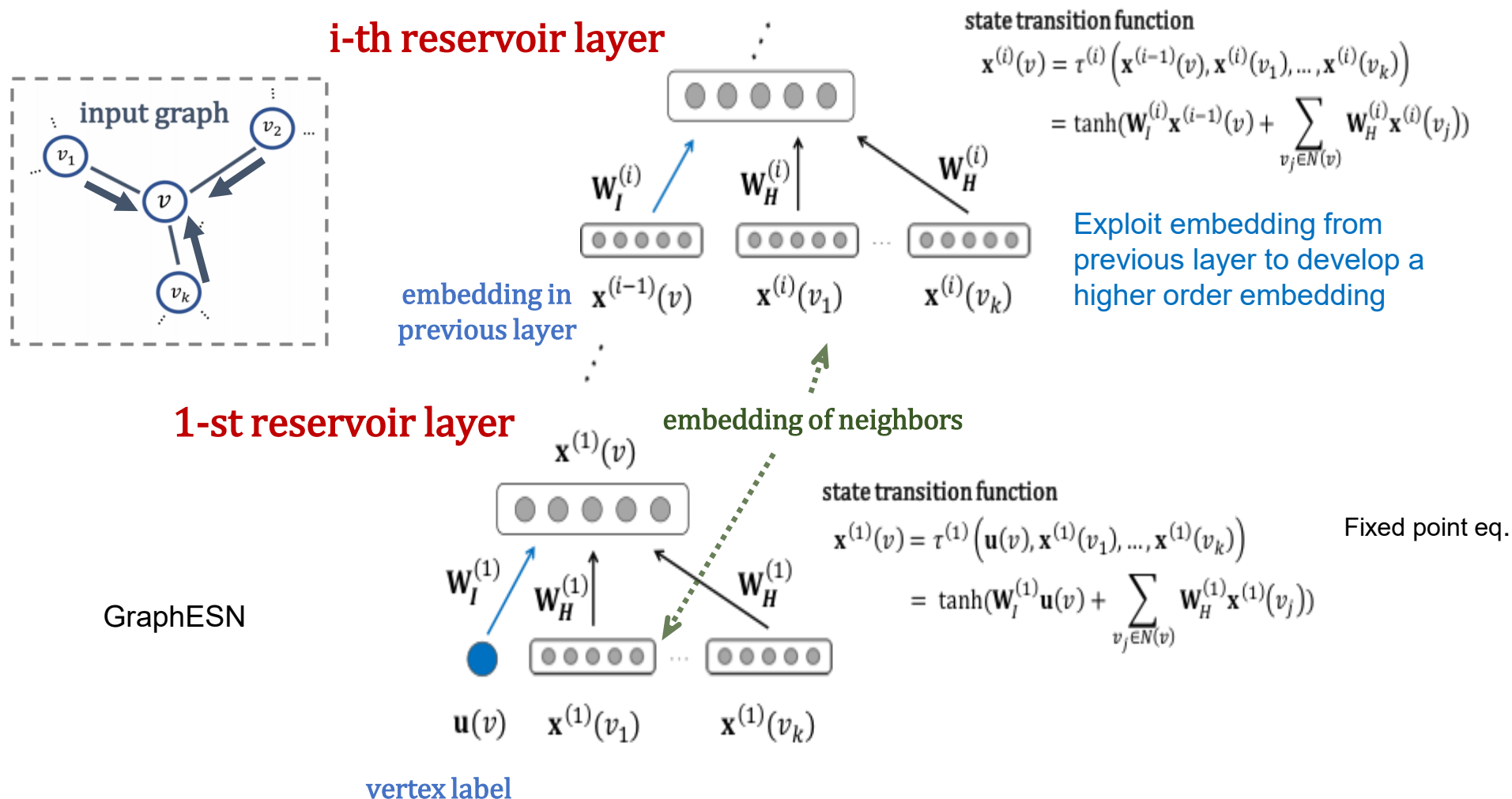
Context evolution, with **the recursive iterations**, of the state, shown for the vertex in the center

Not just local processing, by *diffusion on graph*



1. FDGNN: DeepGESN

Gallicchio, Claudio, and Alessio Micheli. "Fast and Deep Graph Neural Networks." AAAI 2020

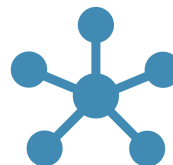


Open Issues in Deep Graph Learning



Efficiency

Scaling training to very large graphs, e.g. >100K nodes



Expressiveness

Deep learning models may fail to learn meaningful node representations



Theory

Characterization of DGN models

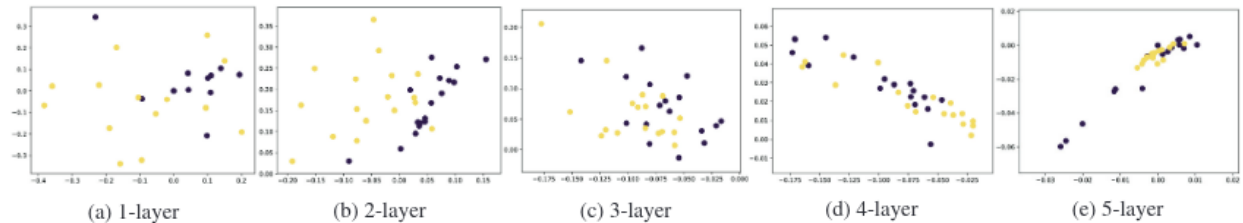
Expressiveness: the Over-* cases



- **Over-smoothing**

- Accuracy typically declines beyond 4-5 layers
- **Node representations collapse** in deeper layers
- Message-passing acts as diffusion
 - *high-level representations become similar*

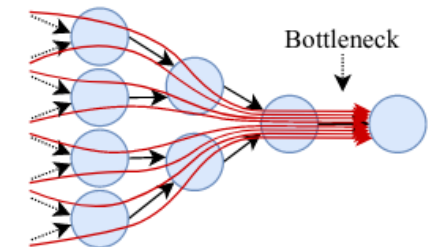
[Li & al. , AAAI 2018]



- **Over-squashing**

- **Bottleneck**: receptive fields increase exponentially with depth in a fixed-size node embedding dimension [Alon, Yahav, ICLR 21].
- Exponential decrease in **sensitivity** of node repres. to the input features, as the number of layers $h_v^{(L)}$ increases, computed as the Jacobian:

$$\left\| \frac{\partial h_v^{(L)}}{\partial x_u} \right\|$$



- **For E2E training**: difficulty of **backpropagating** the gradient through many (MP) layers

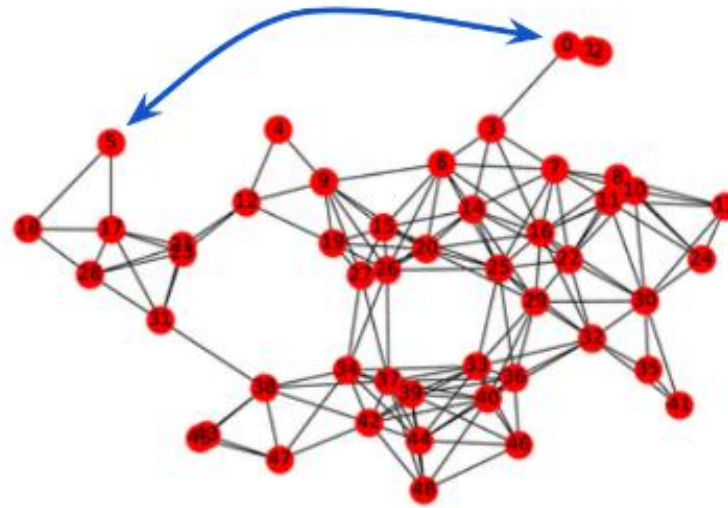
- The **interplay** among them is an open research issue

• Impact 1: Under-reaching



- **Impact 1: Under-reaching**

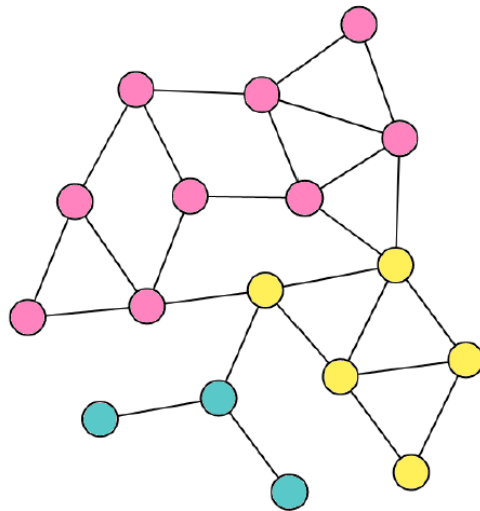
- Over-* may prevent to learn interactions between distant nodes



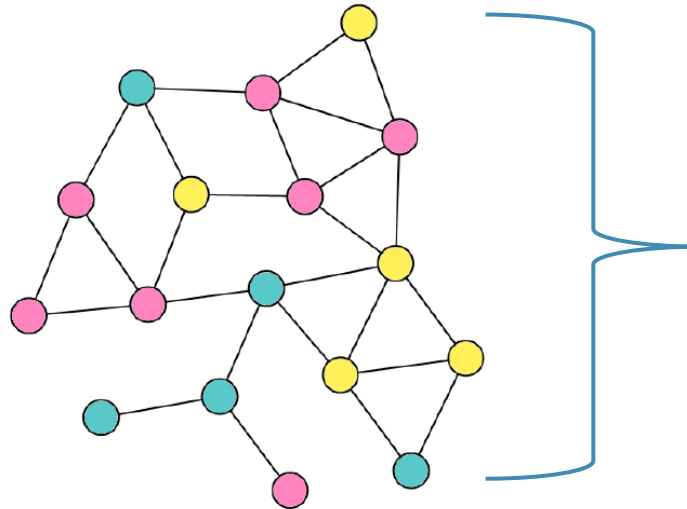
Impact 2: Heterophily Issue in Node Classification



- **Homophilic graphs:** nodes in the same neighborhood mostly share the same class
- **Heterophilic graphs:** nodes belonging to the same class are generally further apart (predictions that rely chiefly on a node's immediate neighbors can be misleading)



high homophily



low homophily

(!)

Potentially **difficult tasks** for standard MP

- impact of the **over-***
- **bias** toward «homogeneous» graphs

A. Micheli, D. Tortorella. *Addressing heterophily in node classification with graph echo state networks*. Neurocomputing (2023).

Theory: Isomorphism and Weisfeiler-Leman (WL) Test



- **How many graphs can DGN learn to distinguish?**

- *Vanilla DGN cannot be more discriminating than WL test, example:*



- **1-dim. Weisfeiler-Leman graph isomorphism heuristic**

- Check a relaxations of graph isomorphism
- Polynomial time heuristic

- **DGNs expressiveness:**

- Message-passing similar to WL coloring
- MP upper bounded by WL *expressiveness*
- There exist versions of DGNs that go beyond 1-WL

- **Characterization of models by bias analysis** → see later

- K. Xu, W. Hu, J. Leskovec, and S. Jegelka, *How Powerful are Graph Neural Networks?* ICLR 2019
- G. A. D'Inverno, M. Bianchini, M. L. Sampoli, and F. Scarselli, *On the approximation capability of GNNs in node classification/regression tasks*, Soft Computing 2024
- C. Morris, M. Ritzert et al. *Weisfeiler and Leman Go Neural: Higher-order Graph Neural Network*, AAAI 2019

Our Proposals



EFFICIENCY ANALYSIS

Beyond E2E backpropagation in DGN

Reservoir Computing for Graphs: Efficiency



Accuracy equal or better than most end-to-end trained models

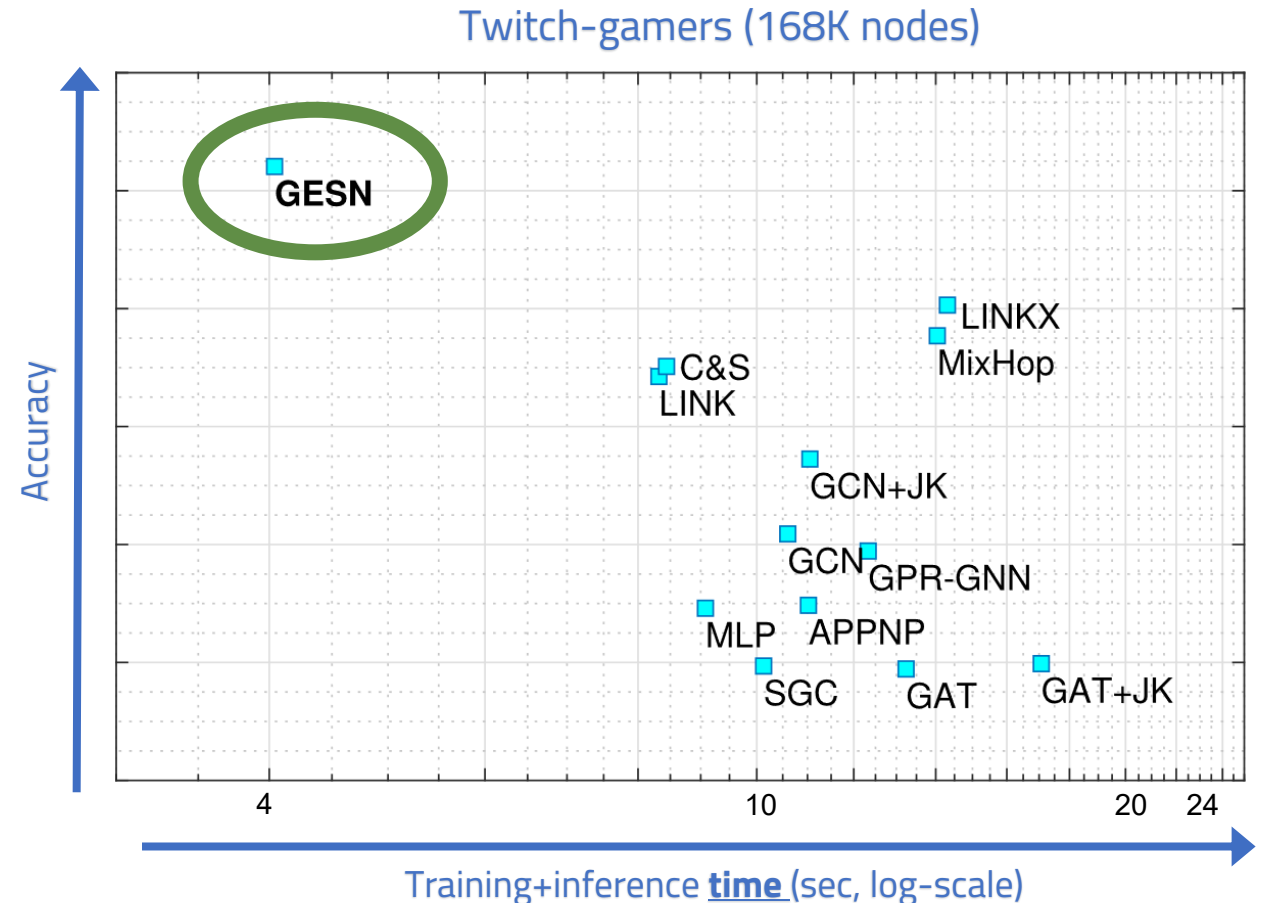
Graph classification:

- GESN is up to **900 times faster**
- Time in **seconds vs hours** even in small data sets...

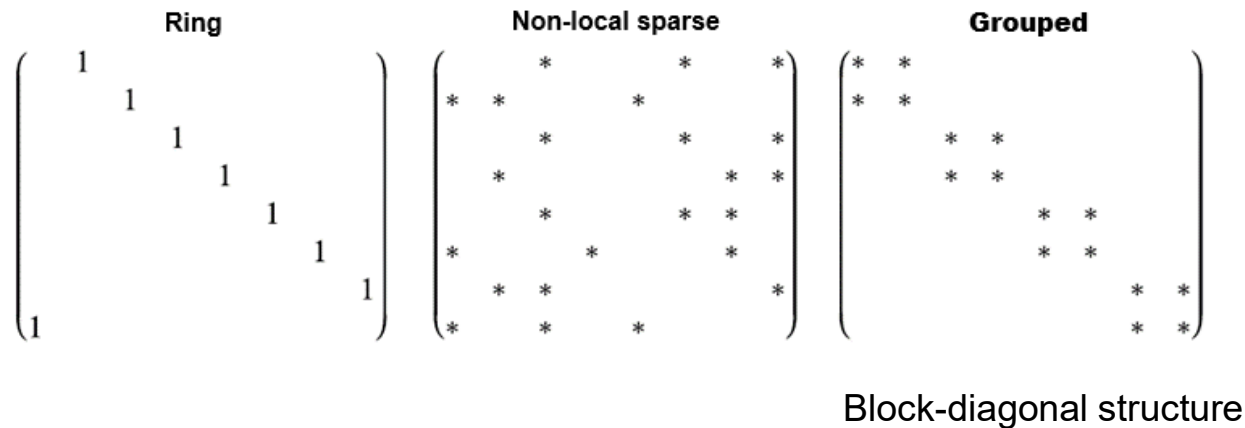
Node classification:

- **A speed-up upto 10x** compared to other models
- Training scales easily to **very large graphs**, e.g. **>100K nodes**

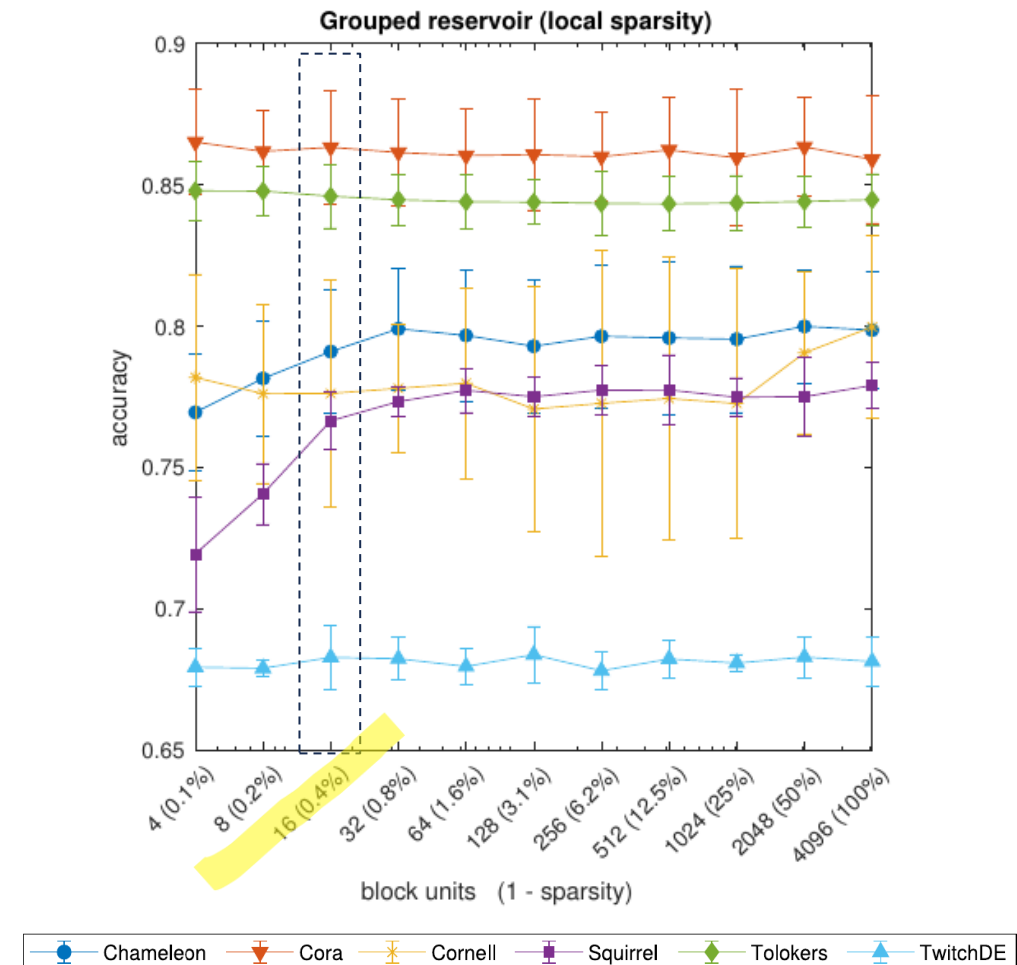
Micheli, Tortorella. *Addressing heterophily with graph echo state networks*. Neurocomputing, 2023



Sparse Design of GESN: Even more efficient



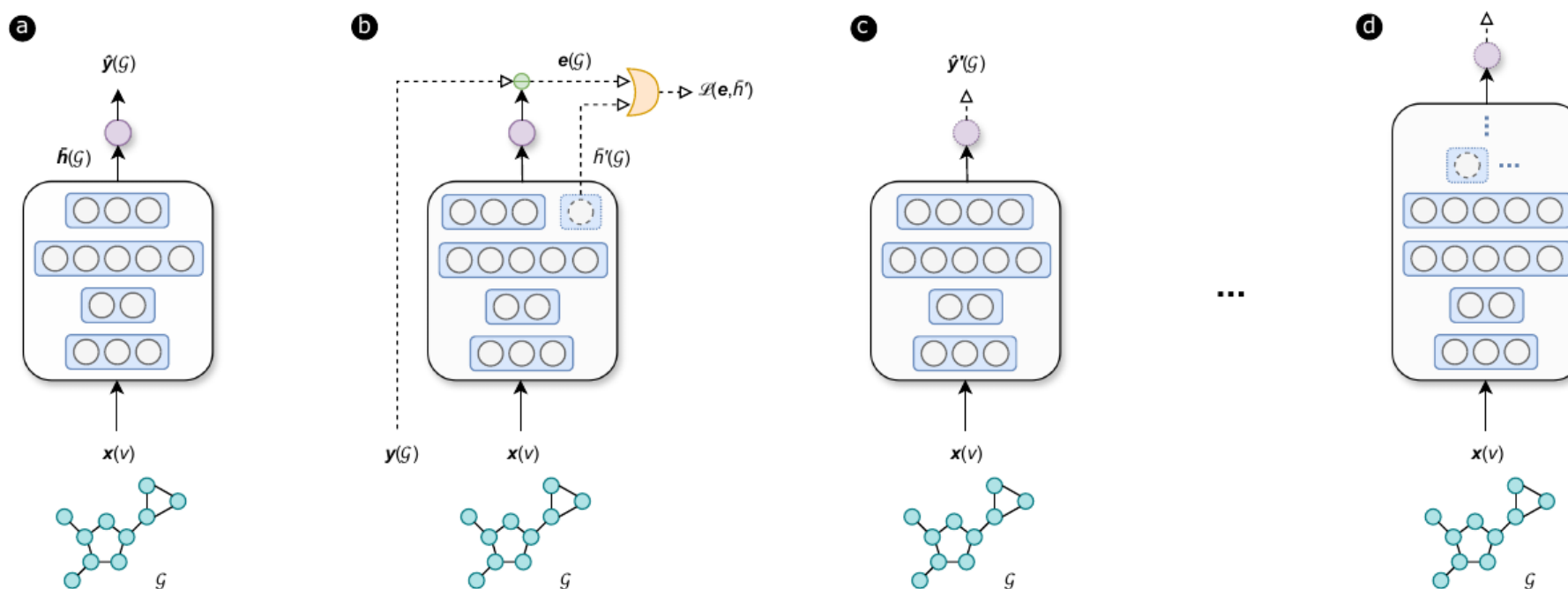
Micheli, Tortorella. *Designs of Graph Echo State Networks for node classification*. Neurocomputing, 2024



Automatic Architecture design by NN4G+ (New)



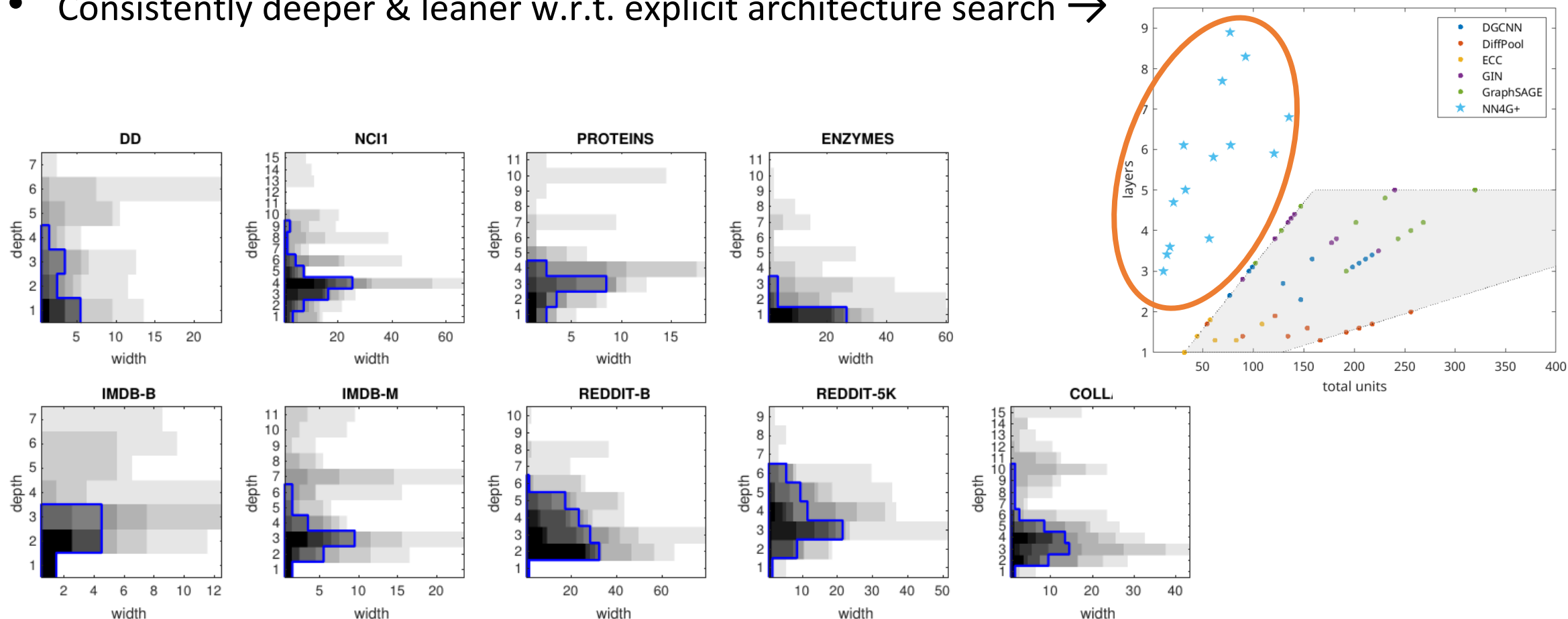
- NN4G+ is built incrementally, with *double synergic expansion* (New!)
 - We train 1 unit at a time: **no E2E backprop** through layers 
 - A layer is greedily expanded until validation accuracy no longer improves
 - New layers are added until validation accuracy no longer improves



NN4G+ Architectures Are Tuned to Tasks (New)



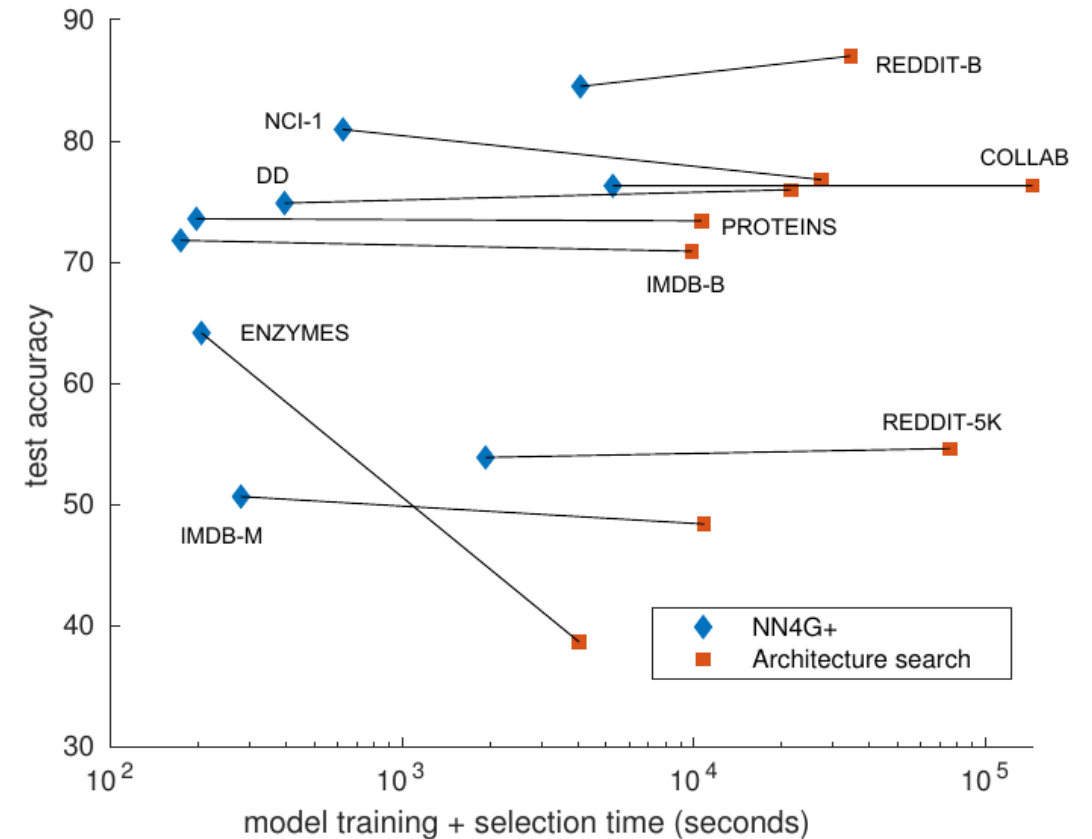
- Consistently deeper & leaner w.r.t. explicit architecture search →

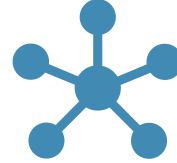


NN4G+ is *Efficient* and Effective (New)



- In general, NN4G+ constructs a neural architecture in *minutes instead of hours* w.r.t. E2E trained models
- In a fair comparison against *explicit architecture selection*, NN4G+ **speeds up >10×** without penalizing accuracy





EXPRESSIVENESS: OVER* ISSUES ANALYSIS

- The **depth** is necessary in DGN but ...
- What are the **impacts of the issues** in Deep Graph Networks?

Expressiveness: Some Tools for the Analysis



- (1) **Models** that *decouple* the MP inductive bias of DGNs from E2E training issues (*going beyond backprop for DGN* 🌱 *)
 - **NN4G**: incremental, *avoiding gradient propagation through layers*
 - **GESN**: *no training* for the embedding, direct control of (fixed) spectral norm $\|\mathbf{W}\|$, imposing $\|\mathbf{W}\| > 1$ to prevent the *sensitivity* w.r.t. the input to be exponentially vanishing (*over-squashing* upper-bound)
- (2) **Looking inside** the models' layers
 - Spectral analysis (graph signals)
 - Accuracy (proxy) ←

$$\text{GCN} \quad \left\| \frac{\partial \mathbf{h}_v^{(L)}}{\partial \mathbf{x}_u} \right\| \leq \left(\prod_{l=1}^L \|\mathbf{W}^{(l)}\| \right) \cdot (\mathbf{A}^L)_{u,v}$$

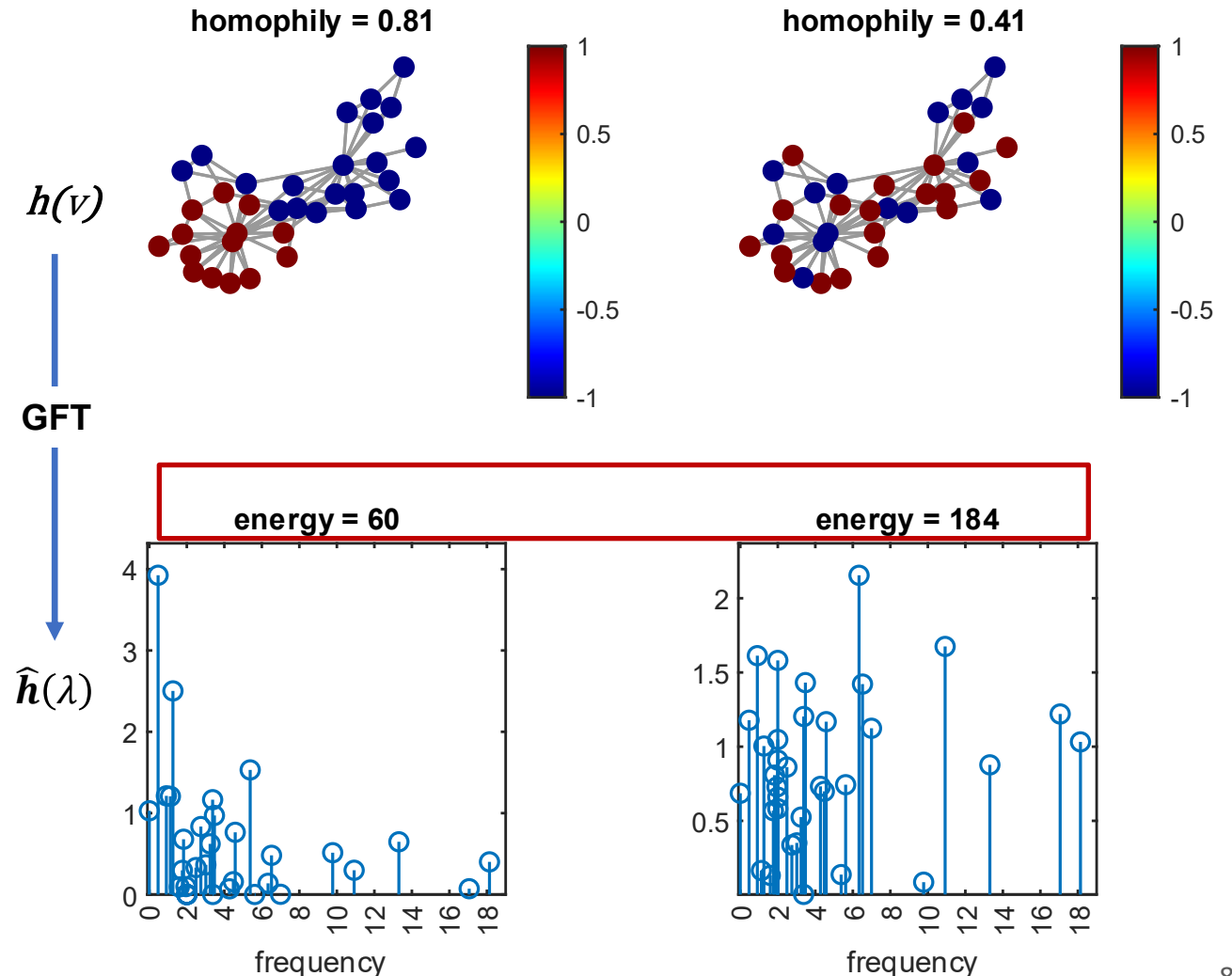
$$\text{GESN} \quad \left\| \frac{\partial \mathbf{h}_v^{(L)}}{\partial \mathbf{x}_u} \right\| \leq \underbrace{\sum_{l=0}^L \|\mathbf{W}\|^l}_{\text{Lipschitz}} \|\mathbf{W}_{in}\| \underbrace{(\mathbf{A}^L)_{u,v}}_{\text{topology}}$$

Spectral Analysis & Etherophilic Graphs (Tech)



- (High) spatial variability of the graph signal, (high) frequency for graphs, are related to low homophily
 - i.e. **larger spectra by Graph Fourier Transform (GFT) for etherophilic graphs** (on the right)
- **Etherophilic tasks require to preserve high frequencies**
- The Dirichelet **energy** $E(\mathbf{h})$ of model layers $\mathbf{h}$ can **measure this**
- Spatial view (embedding variability):
$$E(\mathbf{h}) = \sum_{v \in \mathcal{V}} \sum_{u \in \mathcal{N}_v} \|\mathbf{h}_v - \mathbf{h}_u\|^2$$

- **Spectral view**: *concise measure of the distribution on the graph frequency spectrum*
E.g. high frequency signal → **high energy**



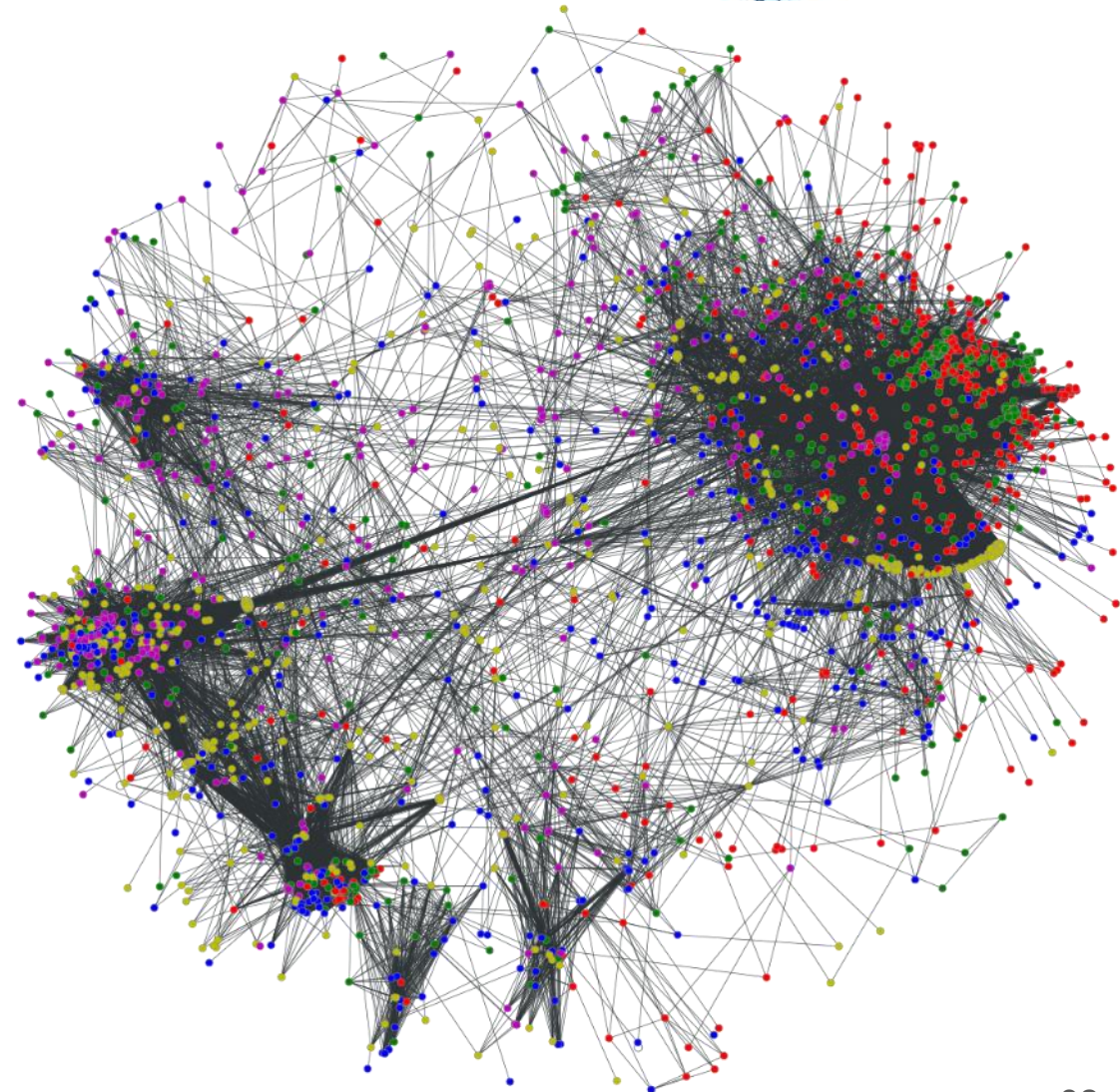
The Task: Chameleon (a «stress test»)



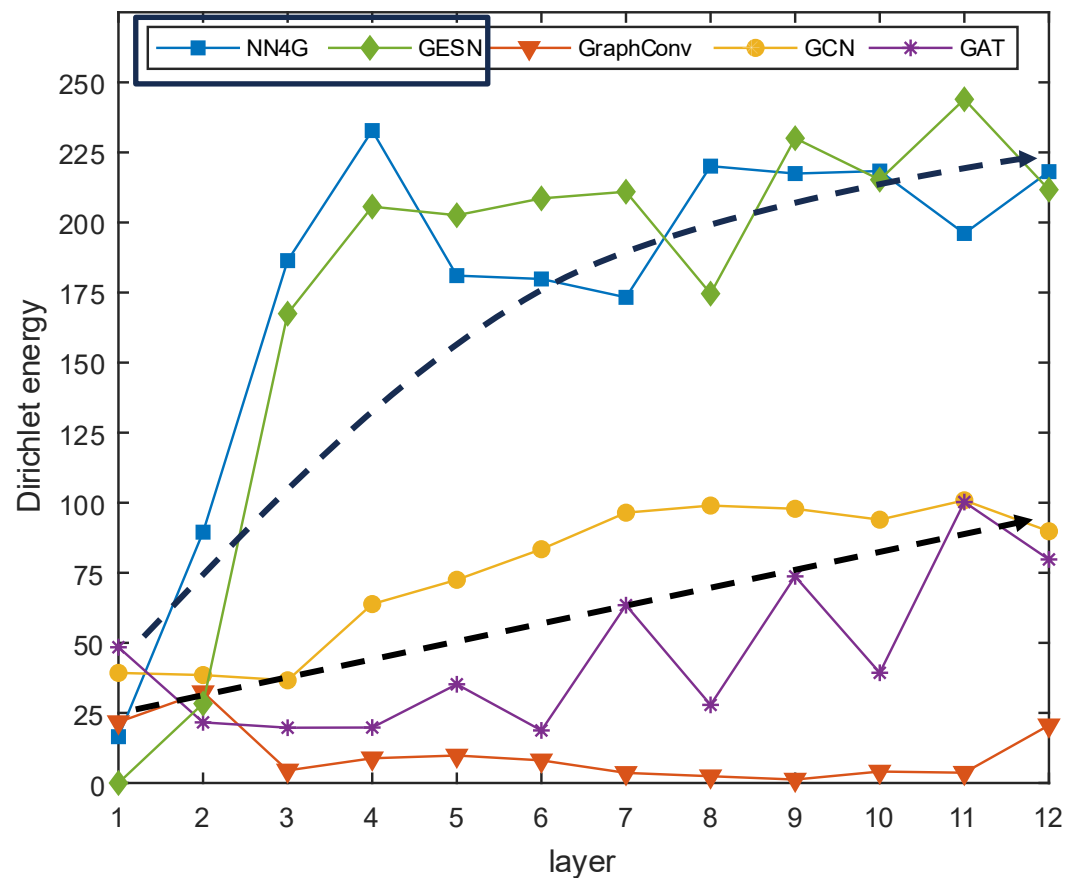
- [Chameleon](#) [Rozemberczki et al, 2019; Pei et al., 2020]
- [Page-page network](#) on a specific topics in Wikipedia
 - nodes represent web pages (bag of words)
 - edges are mutual links between pages
- [5 categories](#) in term of the average monthly traffic of the web page (node classification)
- Nodes: 2277, Edges: 31400
- Degree min/median/mean/max = 1/12/27.6/732
- [Diameter: 11](#) (length of the "longest shortest path")
- [Homophily](#), intra-class edges ratio [0-1]: **0.23**

$$\frac{\text{intra-class edges}}{\text{total edges}}$$

Heterophilic



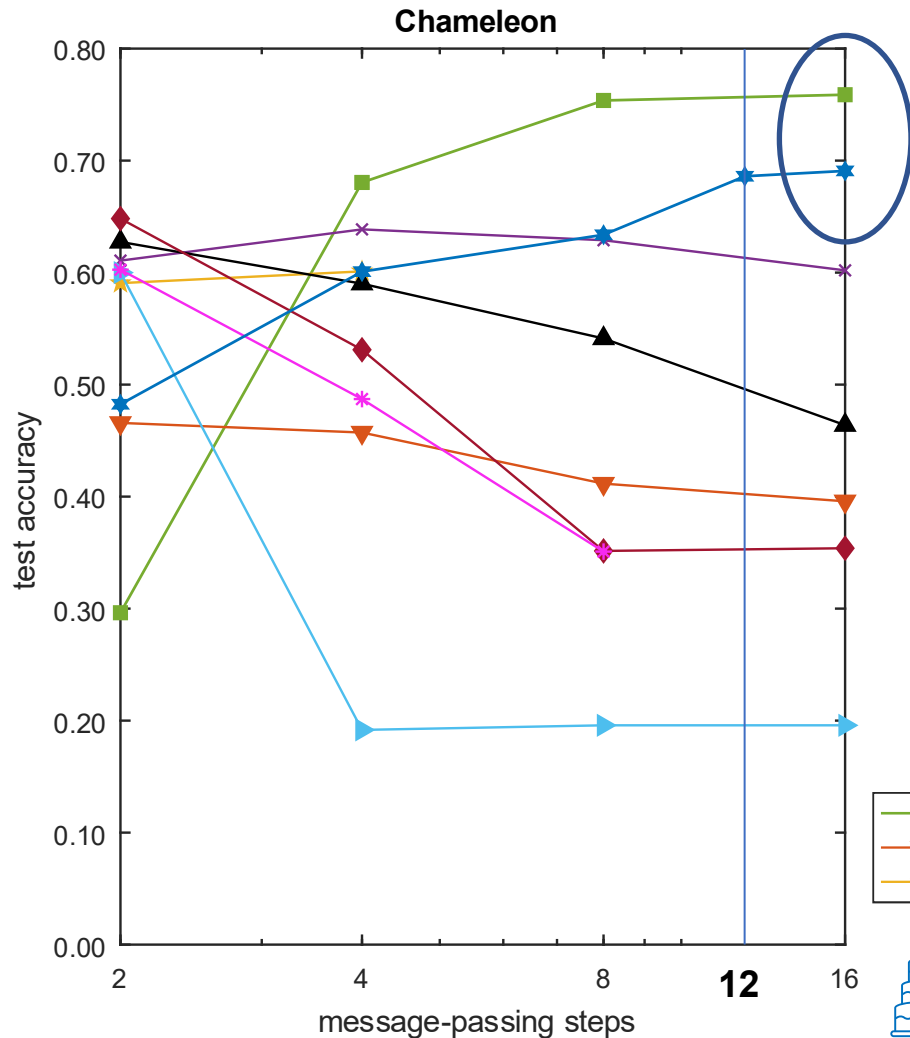
Spectral Analysis (Tech)



NN4G and GESN
better **preserve**
energy through layers
(MP)



Impact of Depth on Accuracy



Accuracy decreases with MP, except for **NN4G** and **GESN**

Ad hoc experiment (Chameleon):

- 16 units per layer (except GESN)
- **12 layers** (to cap the graph diameter)
- Same architectures and units configurations (except GAT that learn edge weights via attention mech.)

Resilience to over-*

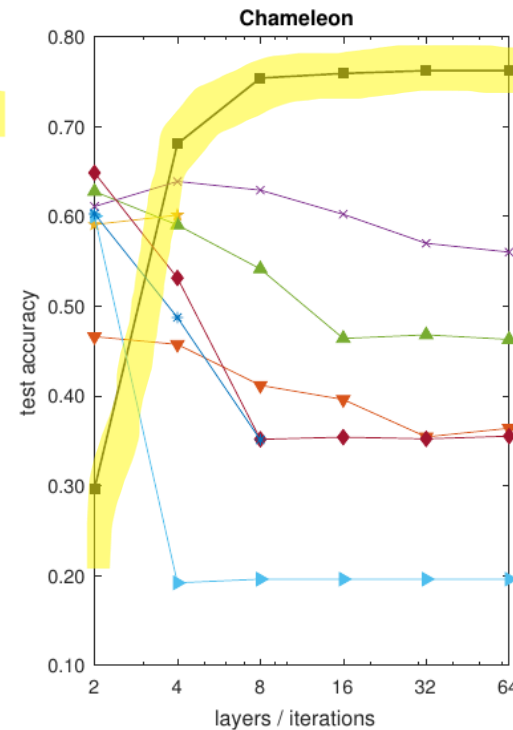
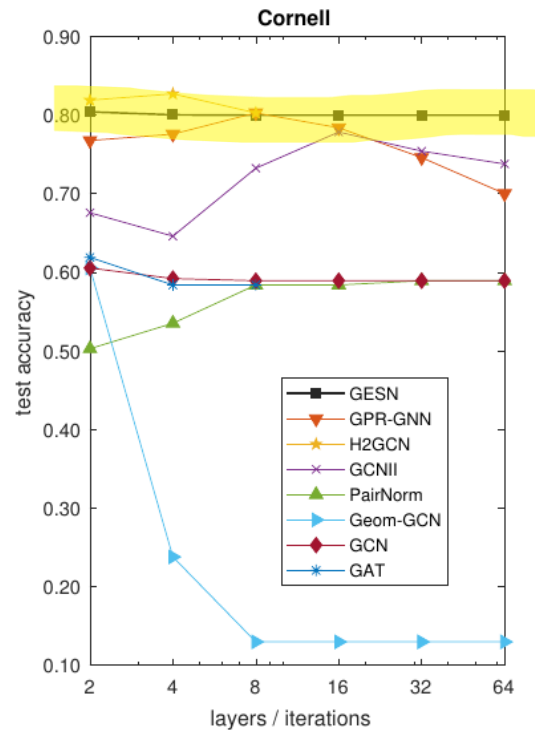
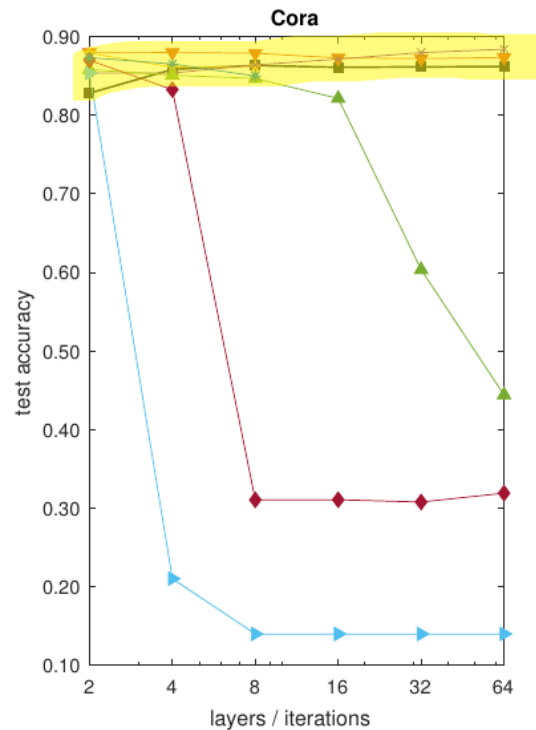
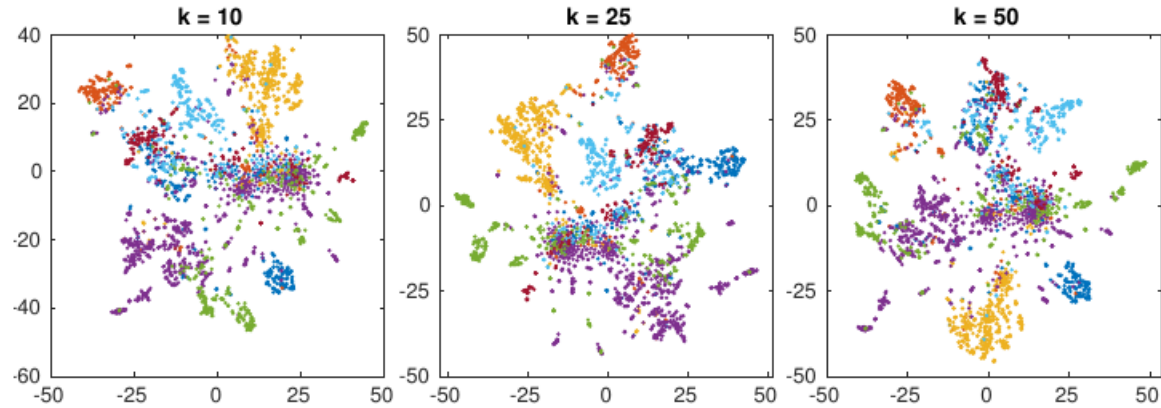
GESN	73.7
NN4G	68.3
GCN	57.8
GAT	55.0
GraphConv	33.8

Results with original models [Yan, Y., et al. ICDM 2022]

Resilience to over-smoothing



Node representations in **GESN** do not collapse as the number of MP steps increases



2D projection of node embeddings for the *Cora* graph at different MP steps. Colors = classes, qualitatively showing how well separable are the node representations.

Micheli, Tortorella. *Addressing heterophily with graph echo state networks*. Neurocomputing (2023)



CHARACTERIZATION OF DGN MODELS

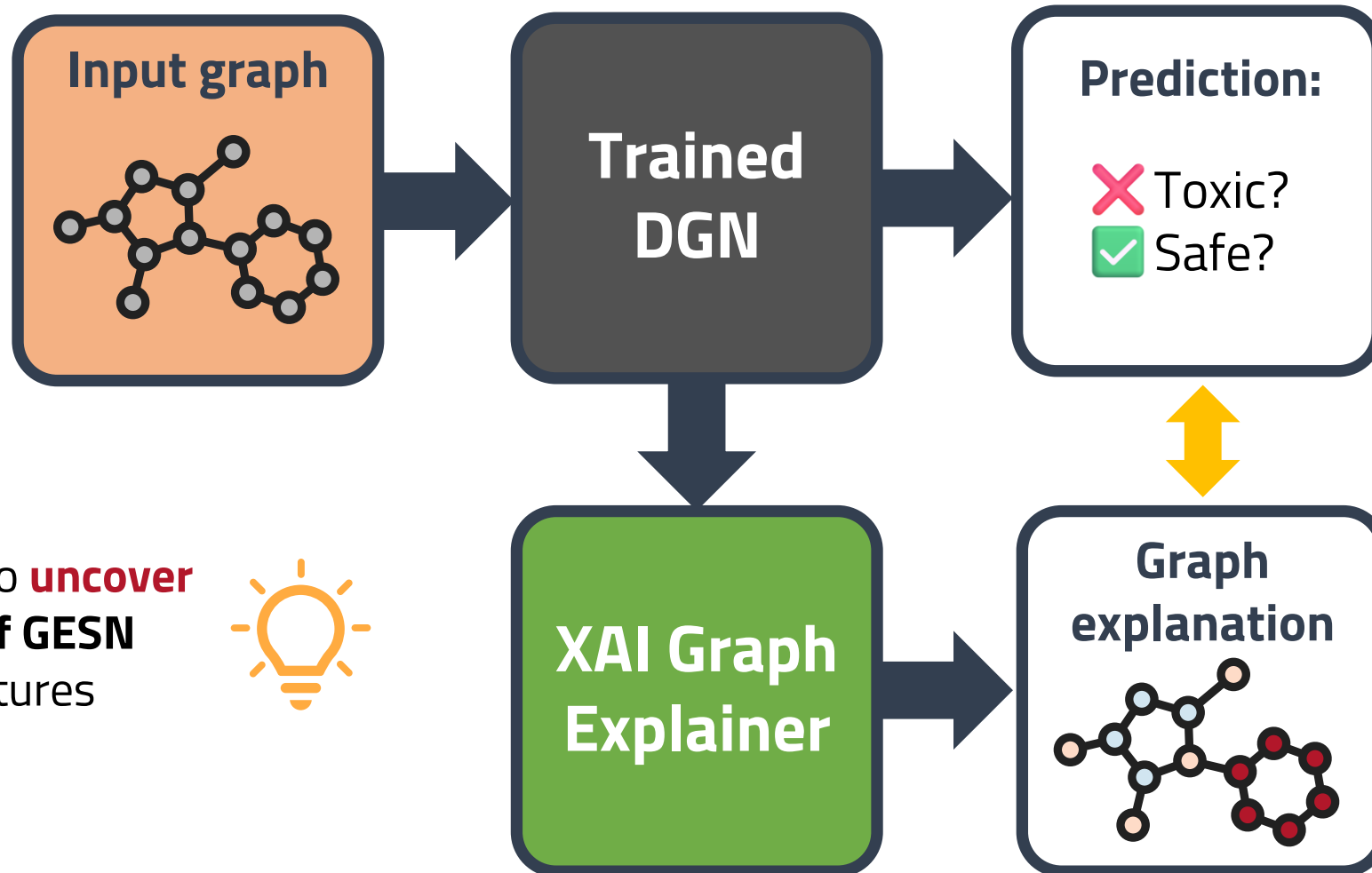
- Further characterization of the models inductive bias
- By the means of **Graph XAI**

Graph XAI as an Investigation Tool



A **graph explainer** derives an importance score for each node of the graph and hence **relevant subgraphs**

We exploited these approaches to **uncover the properties (inductive bias) of GESN** and other different DGN architectures



Explainability 4 Graphs: Our Investigations



With the same level of accuracy (100% for these benchmarks)
GCN and GESN (RC based) show different inductive biases

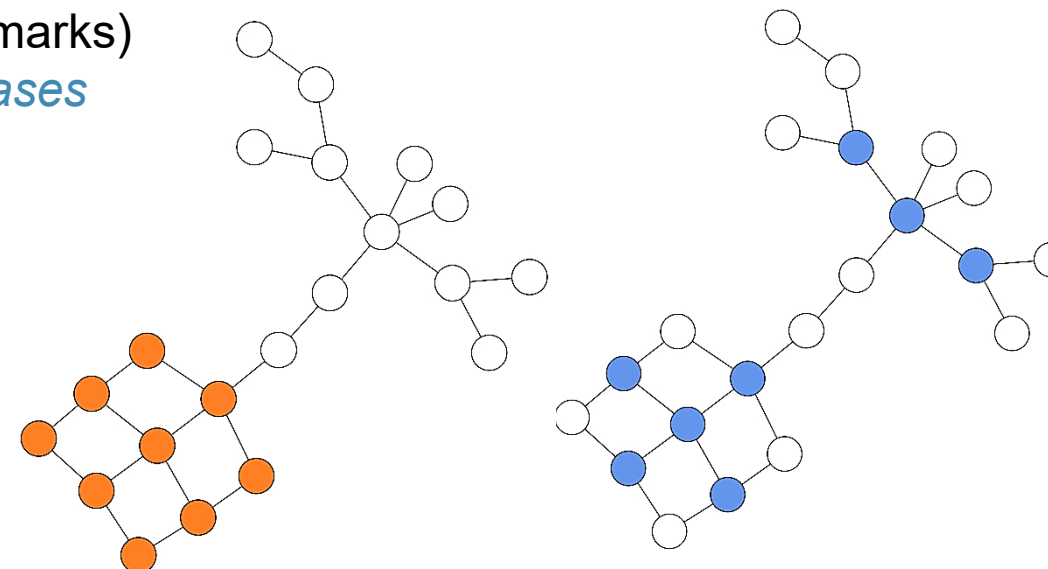
CAM Explainer

Dataset	Motif	
	GESN	GIN
BA2grid	0.806	0.956
BA2Motif	0.769	0.911
GridHouse	0.696	0.903

Dataset	Node degree	
	GESN	GIN
BA2grid	1	0.822
BA2Motif	1	0.812
GridHouse	1	0.847

Plausibility of different DGNs in detecting (different) Ground Truth (GT)

Plausibility measures the alignment (by the AUROC score) between explanation mask and GT



Motif Ground Truth

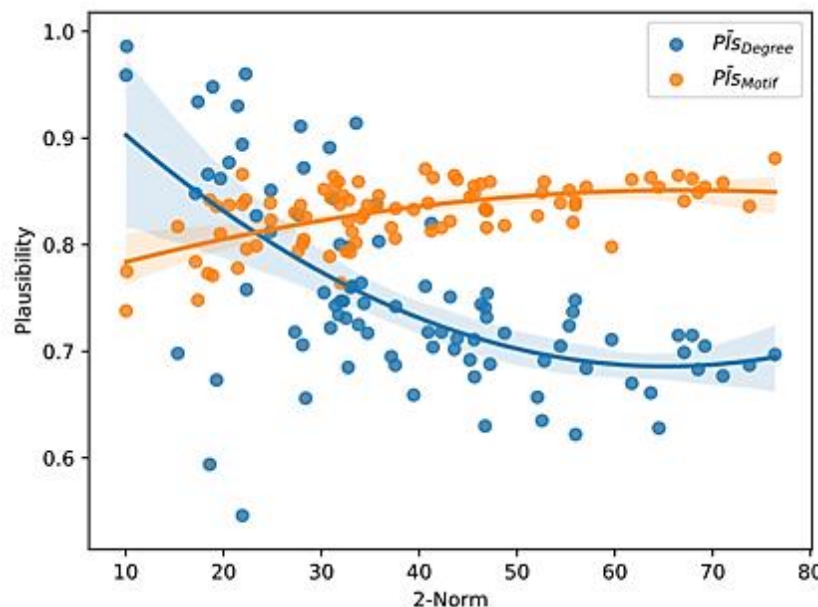
Node Degree Ground Truth

- Some architectures (GCN) prefer detecting **motifs** (left),
- some (GESN) prefer detecting **local degree patterns** (≥ 3) (right)

Explainability 4 Graphs: Our Investigations (II)



- We studied how to correlate to the inductive biases of the models, *also in terms number of layers (**depth**) and weights values*
- As **more layers** are added to a DGN, the detection strategies transit from *degree-first* (low-order structures at the lowest layers) to *motif-first*
- But also the **weights** (training) influences with a shift of explanatory pattern



Plausibility of 2-layered GIN varies as the **weights** 2-norm increases (BA2Motif dataset)

- Node degree pattern
- Motif pattern

→ Richness of capabilities/inductive biases in landscape of DGNs

Explainability 4 Graphs: Our Investigations (III)



The detection policies and *model inductive biases* can be measured and correlated to concepts of **graph theory** (Katz centrality, Fiedler eigenvectors)

Work in progress

- *Real world tasks (e.g. toxicology)*
- *More in-depth analysis of model details*
- *More complex patterns*
- *New benchmarks development ("OpenGraphXAI")*

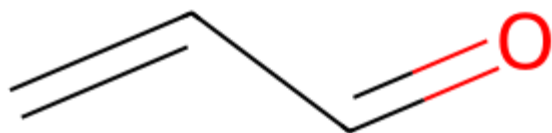


Fontanesi, Micheli, Podda, Tortorella. *Bridging XAI and spectral analysis to investigate the inductive biases of deep graph networks*. Machine Learning, accepted.

Applications: example on Tox21 P53



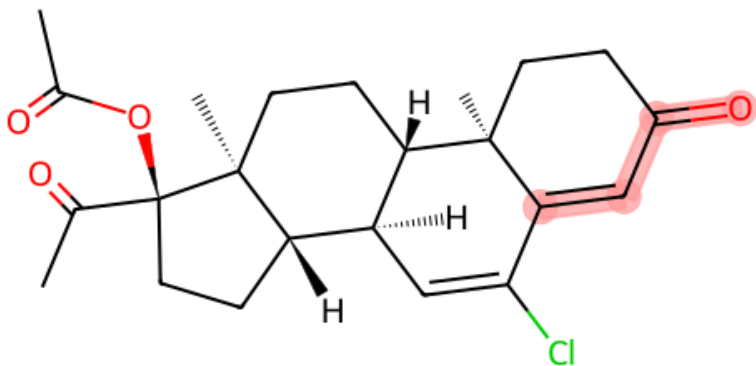
Michael acceptor



XAI: Identify the (known) **fragment(s)** (structural patterns) that influences the activation

Structural Pattern: **Michael acceptor**

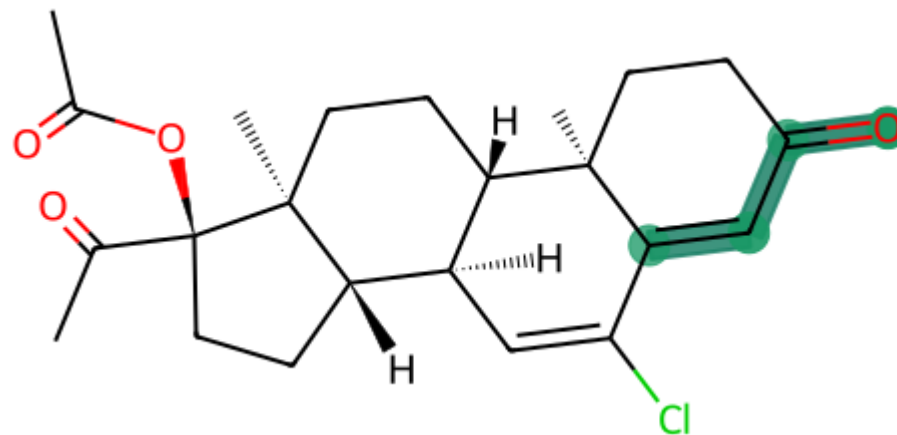
Smart: "[C]=[C]-[C](=[O])"



Name: chlormadinone acetate

Smile: [H][C@@]12CC[C@](OC(C)=O)(C(C)=O)[C@@]1(C)CC[C@@]3([H])[C@@]2([H])C=C(Cl)C4=CC(=O)CC[C@]34C

XAI Result



Top4 Nodes

Graph XAI: Emerging Messages



- Relevance to **know your models** (which bias you want to exploit during learning)
- **The truth is not unique!** There exist **multiple valid solutions** (*same classification accuracy*) from different models but having *different explanatory patterns*
- This is particularly interesting **for graphs**, where there is **not** the same “natural” visual interpretation that we can have on images
- The **GT is rarely available** in real-world contexts **for graphs**: It is extremely necessary to introduce accurate annotated benchmarks for **graph XAI** from real data
- We proposed **OpenGraph XAI** with more than 15 novel datasets derived from molecular datasets and a software framework able to generate **2000+ XAI benchmarks** with controlled GT

Fontanesi, M., Micheli, A., Podda, M., Tortorella, D.: Addressing the scarcity of benchmarks for graph xai (2025), <https://arxiv.org/abs/2505>.



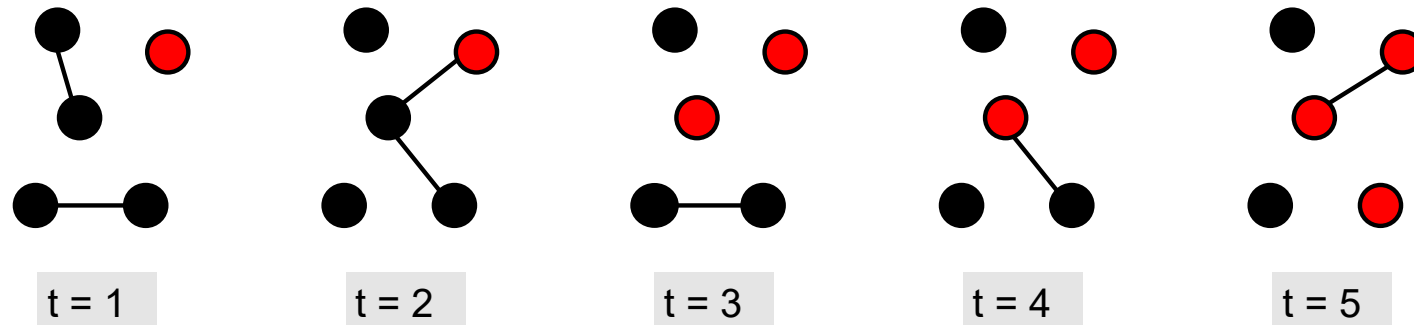


EXTENSIONS

Scope: Dynamical/temporal graphs



- Dynamic/Temporal Graphs represent relationships or node values that evolve through time
 - E.g. Infection-spreading models, content dissemination on social networks, signals on nodes...



A. Micheli, D. Tortorella. *Discrete-time Dynamic Graph Echo State Networks*. Neurocomputing (2022)

Dynamic Graph Echo State networks: Results

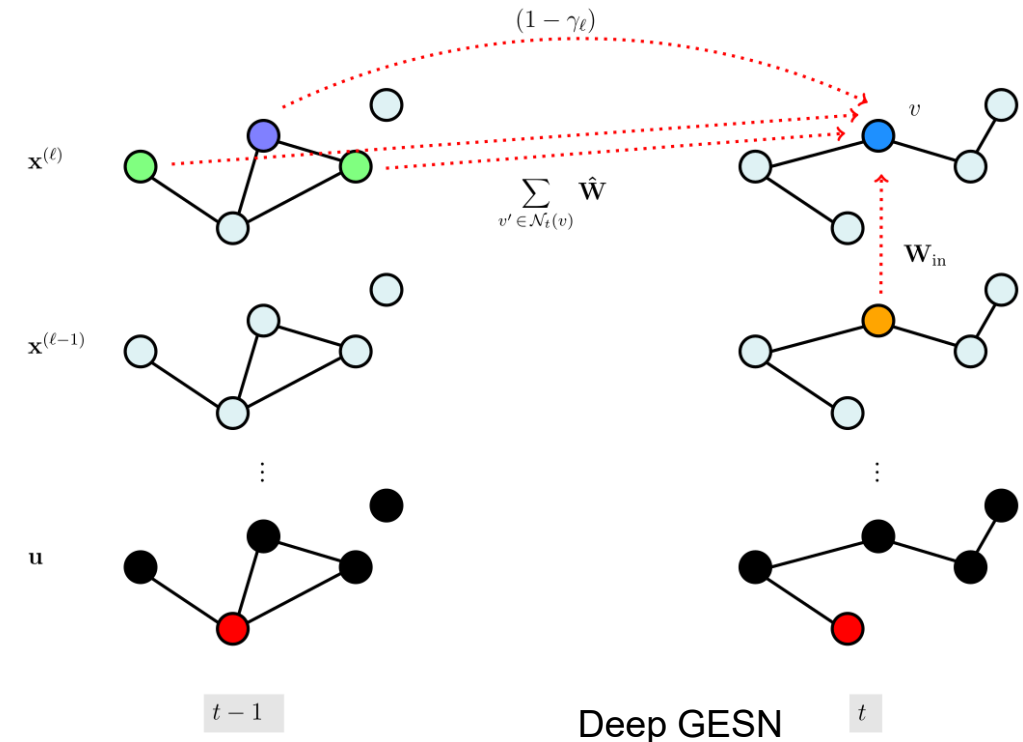


- Multi-layer reservoir defined by the node-wise state transition function, implementing a spatio-temporal graph convolution:

$$\mathbf{x}_t^{(\ell)}(v) = \gamma_\ell \tanh \left(\mathbf{W}_{\text{in}}^{(\ell)} \mathbf{x}_t^{(\ell-1)}(v) + \sum_{v' \in \mathcal{N}_t(v)} \hat{\mathbf{W}}^{(\ell)} \mathbf{x}_{t-1}^{(\ell)}(v') \right) + (1 - \gamma_\ell) \mathbf{x}_{t-1}^{(\ell)}(v)$$

- Stability condition:** $\|\hat{\mathbf{W}}^{(\ell)}\| < \frac{1}{\|\mathbf{A}_t\|}$ for all layers (time-dependent, geometric mean of the graph series)

- More or as effective as (10) TGNs (fully-trained Temporal Graph neural Nets) but ...
 - 100x** training speed-up w.r.t. TGNs
 - 10x** inference speed-up w.r.t. TGNs



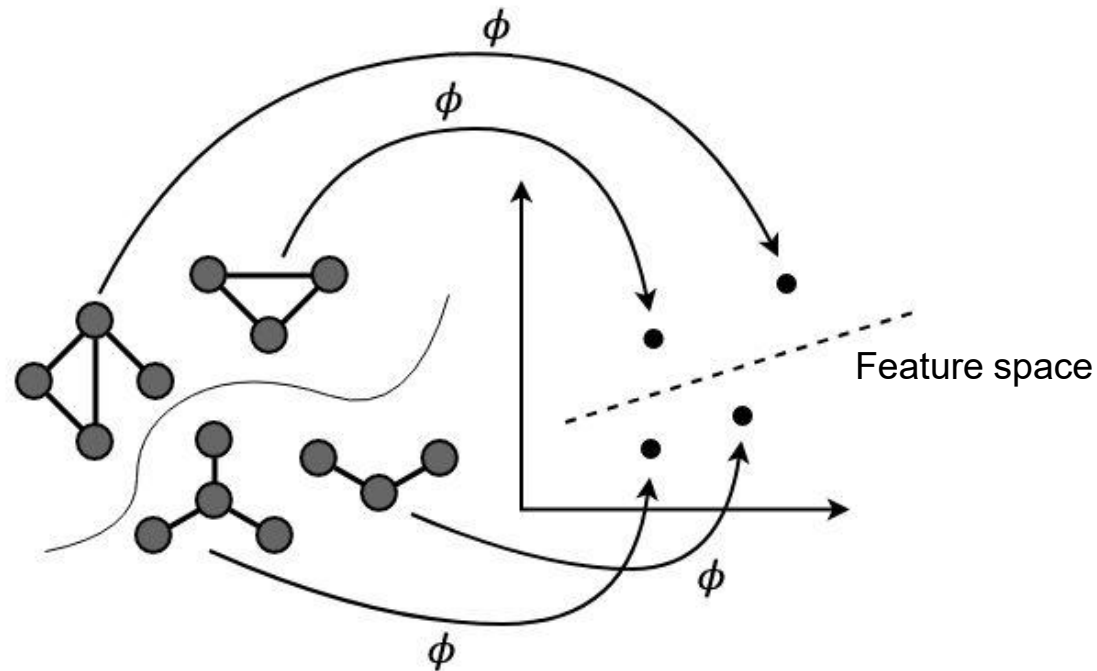
Micheli, Tortorella. *Discrete-time Dynamic Graph Echo State Networks*. Neurocomputing (2022).

Other approaches to Learn in SD: Example

And Kernels?



- Large set of proposals (out of the scope)



Kernel methods for SD



Kernel modularity/ definition: easily can be extended to graphs of different types: sequences, trees, **graphs** (directed, undirected, cyclic/acyclic, ...)

Implicit embedding of data into Euclidean space by specifying an inner product

$$k(x, x') = \langle \phi(x), \phi(x') \rangle$$

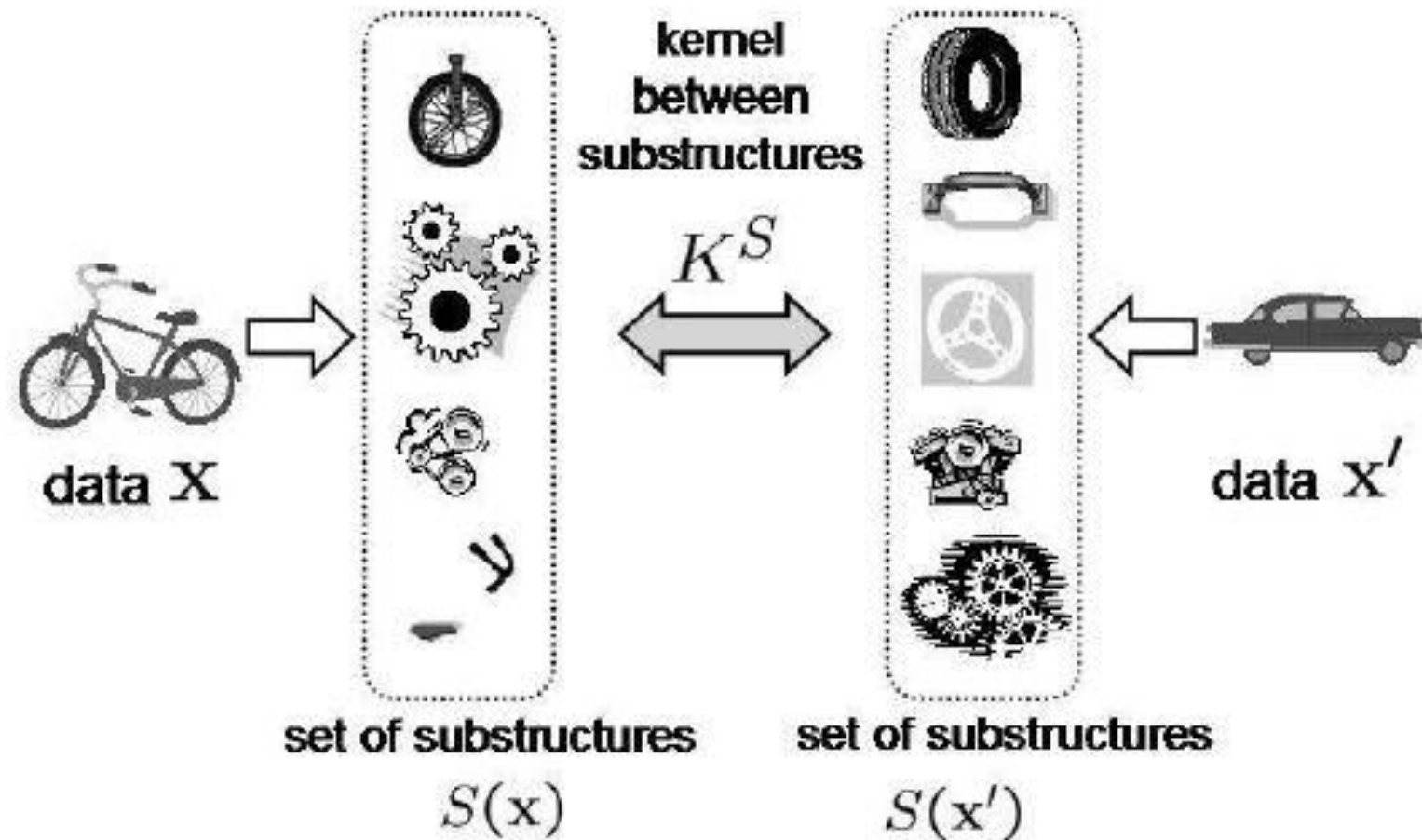
Data $\mathbf{x}$ can be of various types, including structured data

Define similarity between data

Example: Marginalized Kernel

- Similarity between 2 graphs based on the *common label paths* obtained by *random walks* in the graphs
- Kernel function defined as the inner product of the count vectors, averaged over all possible label paths
- Computational complexity of the kernel function (typically) *scales quadratically with the graph size*

Convolution Kernels



Kernels for SD: discussion



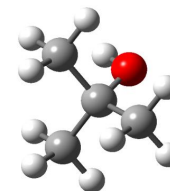
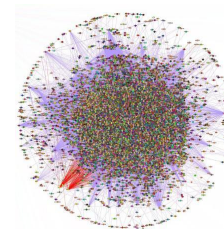
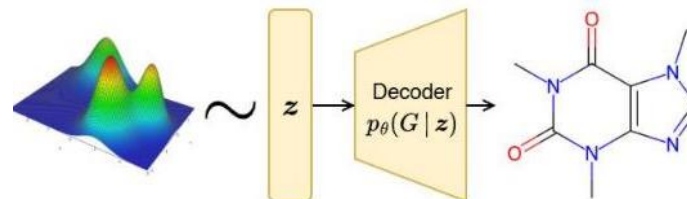
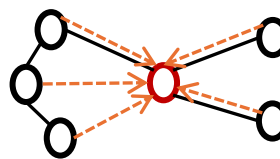
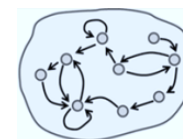
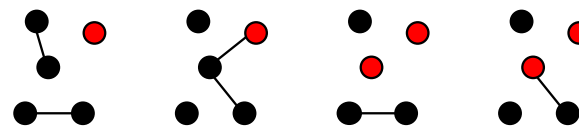
- Efficiency: the cost is critical (due to sub-structure decomposition)
- **Adaptivity warning:** Kernel for structure: *knowledge-driven methods*
- Similarity measure fixed prior to learning
 - i.e. no adaptivity/learning in the encoding into the feature space
 - **Pro:** we can exploit prior knowledge by the Kernel (to engineering a specific problem)
 - **Cons:** For SD ***no universality is known:***
 - The choice of the kernel function is critical
 - E.g. Computing any complete graph kernel *is at least as hard as* **'Graph Isomorphism'**
- Adaptive kernels studies...

Perspectives & Conclusions

Perspectives (Examples)



- **Scope and extensions:** Dynamics graphs, heterogenous graphs, hypergraphs, etc.
- Evolution **RC** based models
- **Graph Transformers** (attention mechanism considering all the nodes)
 - similarities with MP, many variants, not always better performances, efficiency issue ($O(n^2)$)
- **Graphs generation**
- **Foundation models**
- **XAI** can help to **uncover** properties of DGN and to properly use them, but we are just at the beginning
- **Trustworthy** concepts
- **Applications**, e.g. Biological networks, PPI; Cheminformatics, ...



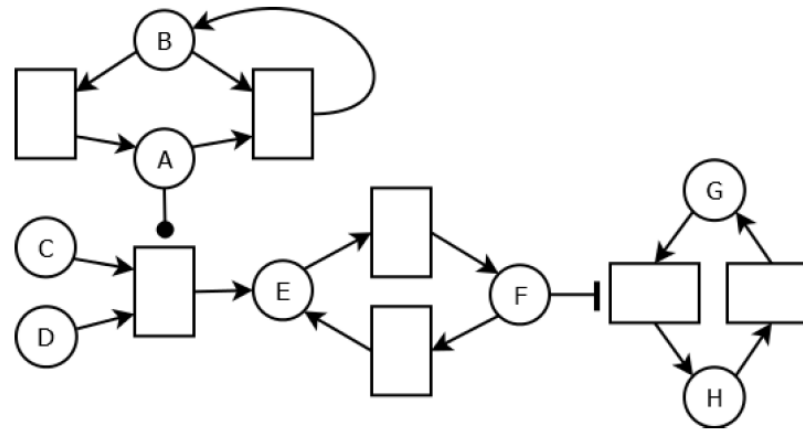
Applications (Example 1)



Biochemical Pathways (BPs) are sets of chemical reactions among biomolecules
They model **cell functionality**

Structure described by **graphs** (e.g. **Petri Nets**)

Goal: predict dynamical properties associated to the pathways (robustness, sensitivity, etc.)



Fontanesi, M., Micheli, A., Milazzo, P., & Podda, M. (2023). Exploiting the structure of biochemical pathways to investigate dynamical properties with neural networks for graphs. *Bioinformatics*, 39(11)

Applications (Example 2)

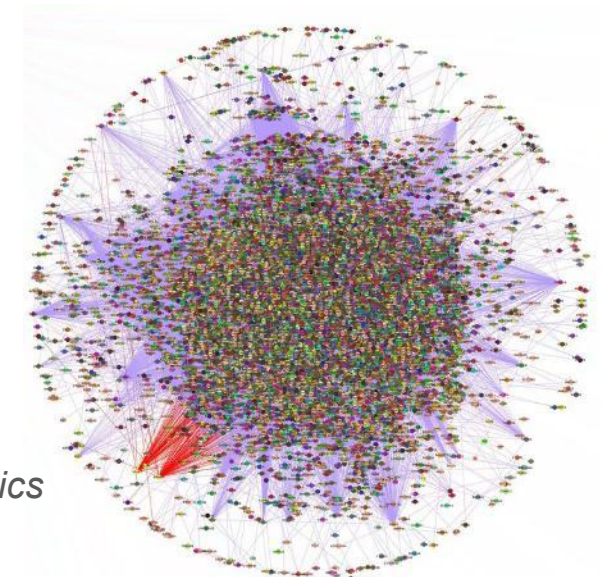


- Chemical, social networks benchmarks
- **Biology:**
 - Protein-Protein Interaction Network (**PPI**)
 - *Nodes* represent proteins
 - *Arcs* represent physical associations between two proteins
 - **Predicting sensitivity** (perturbation analysis) on PPI
 - Starting from biochemical pathways (Biomodels) and mapping to PPI subgraphs (BioGRID, **60K nodes**)
- **Results:**
 - **Efficiency** vs numerical simulation on biochemical pathways (ODE)
 - **High performance** even without protein embeddings, meaning that DGN is relying on PPI topology
- **Potential usage:**
 - Disease understanding
 - Drug design and repurposing
 - Precision/Personalized medicine



PAN-HUB prj on
antibiotics resistance

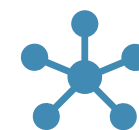
Dipalma, et al. *BMC Bioinformatics*
(2025)



Conclusions



- It is possible to do **deep learning** for **graphs** **efficiently**
(without fully E2E training)
 - Models as GESN and NN4G allow to efficiently build DGN with high accuracy
 - At the same time enabling the investigation of the MP biases
- ML for **graphs** is a concrete reality
 - It can provide new solutions for complex domains where the **managing of relationships** is relevant to achieve suitable modeling of the solutions
 - If you have data with relationships (serial, hierarchical, networks), any domain where you find structured data, **do not use a flat representation** ... we can work together on them!
 - While still **under development**



General conclusions



Many open issues & extensions



Many applicative fields

We are open to collaborations!
Your Thesis and more  *“tie a knot in your handkerchief”*

CIML group @ C.S. Dep. - University of Pisa



- **Faculty** (9): Prof. A. Micheli (coordinator), Prof. D. Bacciu, Prof. C. Gallicchio, Dr. A. Carta, Dr. A. Ceni, Dr. A. Cossu, Dr. L. Passaro, Dr. M. Podda, Dr. D. Tortorella
- 5+ post-doc researchers
- **12+ PhD students (the young energy!)**
- More than **25 years** of research in ML/AI
- Part of the cairne.eu Research Network **CAIRNE**
(Confederation of Laboratories for Artificial Intelligence Research in Europe)

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**Computational Intelligence &
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Special Thanks



- **Claudio Gallicchio**, Assoc. Prof.
- **Marco Podda**, Researcher (FAIR)
- **Luca Pedrelli**, Post-Doc (THE)
- **Domenico Tortorella**, Researcher (PAN-HUB)
- **Michele Fontanesi**, PhD student (FAIR)
- **Alessandro Dipalma**, PhD student



Projects

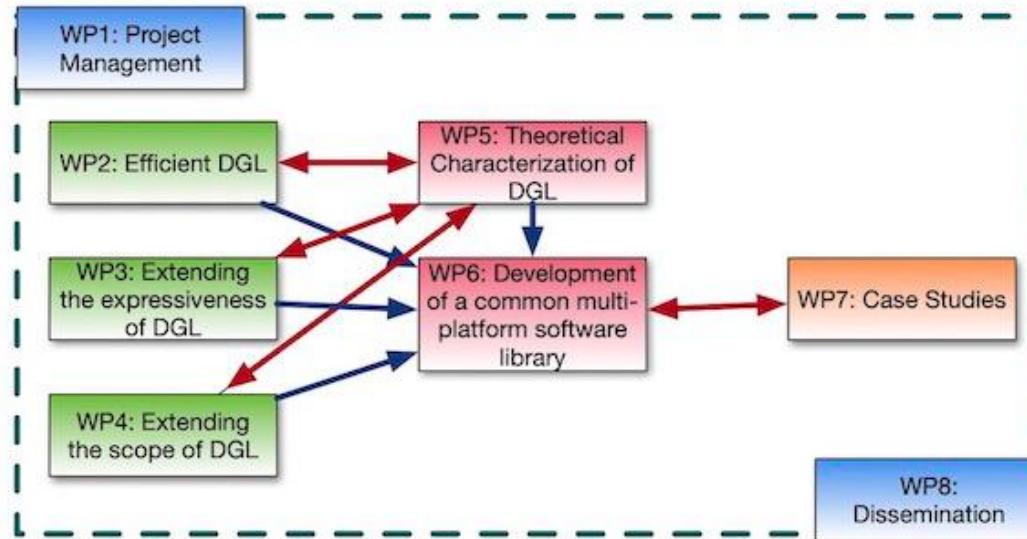
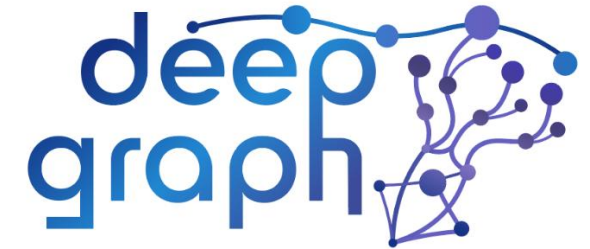
- Research for extension of **DGN** to **continual learning, automatic model design** within the **trustworthy** concepts (*explainability*, robustness, fairness, privacy) is open....
(graphs are discrete structures!)
- PNRR PE00000013 "FAIR - Future Artificial Intelligence Research" - Spoke 1 "**Human-centered AI**", funded by the European Commission under the *NextGeneration EU programme*. 2023, 3 years
- WP on *Human centered Lifelong Learning for **Complex Data***



Projects



- PRIN (national project), 2Y, starting Feb. 2025
- **DEEP-GRAPH: Design and Theory of Deep Graph Learning**
 - University of Padova (Prof. A. Sperduti)
 - University of Pisa (Prof. A. Micheli)
 - University of Siena (Prof. F. Scarselli)



Ministero
dell'Università
e della Ricerca



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Thank You



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